NM-W3-Bodorin-v2

Evaluating functions by continued fractions

Solving transcendental equations by Rolle's sequence method

Numerical Methods · theory, algorithms and 10 fully worked examples

Roadmap of the presentation

I Evaluating functions by continued fractions

- Definition, notation, convergents
- Generating a CF from scratch: successive reciprocals
- Tools: series reciprocal, equivalence transformation
- Evaluation: backward scheme, forward recurrence
- 5 examples, each generated then evaluated: e^x, tan x, ln(1+x), arctan x, √10

II Transcendental equations: Rolle's sequence method

- Rolle's theorem and its corollary
- Building Rolle's sequence, counting sign changes
- Separating every real root
- Refining each root by bisection
- 5 worked examples incl. a two-level Rolle sequence

Part I · Evaluating functions by continued fractions

Replace an infinite expansion by a short chain of divisions — and get many correct digits fast

What is a continued fraction?

General (staircase) form

$$f(x) = b_0 + \cfrac{a_1}{b_1 + \cfrac{a_2}{b_2 + \cfrac{a_3}{b_3 + \cfrac{a_4}{b_4 + \cdots}}}}$$

Compact notation: $f = b_0 + a_1/(b_1+)\ a_2/(b_2+)\ a_3/(b_3+)\ \cdots$

$\mathbf{a_k}$	partial numerators (may depend on x)
$\mathbf{b_k}$	partial denominators (may depend on x)
$\mathbf{C_n}$	n-th convergent: cut the fraction after a_n/b_n
$\mathbf{C_n \to f}$	convergence: the approximation improves with n

Idea: evaluating f(x) = computing a convergent $C_n(x)$, a rational function obtained with only n divisions.

Generating a continued fraction: the one identity behind it

$$f = \beta + (f - \beta) = \beta + \alpha x^m / f_1 , \quad \text{where} \quad f_1 = \alpha x^m / (f - \beta)$$

After step 0

$$f = \beta_0 + \cfrac{\alpha_1 x^m}{f_1}$$

After step 1

$$f = \beta_0 + \cfrac{\alpha_1 x^m}{1 + \cfrac{\alpha_2 x^m}{f_2}}$$

After step 2

$$f = \beta_0 + \cfrac{\alpha_1 x^m}{1 + \cfrac{\alpha_2 x^m}{1 + \cfrac{\alpha_3 x^m}{f_3}}}$$

Read it as a recipe: take off the constant term β, keep the lowest power αx^m of what is left as the next numerator, and continue with the reciprocal f_1. The tail f_k is the only unknown; replacing the last tail by 1 gives the approximant C_n.

The successive-reciprocal algorithm (Viskovatov), step by step

1 Start — $f_0(x) = f(x) = c_0 + c_1x + c_2x^2 + \ldots$ (Taylor series, about n + 1 terms)

2 Constant term — $\beta_k = f_k(0)$

3 Lowest term of the rest — $f_k - \beta_k = \alpha_{k+1}x^m \cdot (1 + d_1x + d_2x^2 + \ldots) = \alpha_{k+1}x^m \cdot q_k$

4 Reciprocal (Tool 1) — $f_{k+1} = 1/q_k = 1 + e_1x + e_2x^2 + \ldots$ then $k \leftarrow k + 1$

repeat n times

5 Assemble after n steps — $f \approx \beta_0 + \alpha_1x^m/(1 + \alpha_2x^m/(1 + \cdots + \alpha_nx^m/1))$

6 Clean up (Tool 2) — multiply level k by $\lambda_k \rightarrow$ integer a_k, b_k; read the pattern

Bookkeeping

- $\beta_k = 1$ for every $k \geq 1$, because $1/q_k$ starts with 1.
- If f(0) = 0 (tan, ln(1+x), arctan) then $\beta_0 = 0$.
- m = 1 normally; m = 2 for odd/even functions (series in x^2).
- Each step uses up m terms of the series, so keep every series only as long as it is still needed.

Tool 1: the reciprocal of a power series, by hand

Problem

$q = 1 + d_1x + d_2x^2 + d_3x^3 + \ldots$ find $1/q = e_0 + e_1x + e_2x^2 + \ldots$

Idea

$q \cdot (1/q) = 1 \Rightarrow$ coefficient of x^k $(k \geq 1)$ must vanish:

$e_k + d_1e_{k-1} + d_2e_{k-2} + \ldots + d_ke_0 = 0$

$$\mathbf{e_0 = 1, \quad e_k = -(d_1e_{k-1} + d_2e_{k-2} + \ldots + d_ke_0)}$$

Worked mini-example: $q = 1 + 2x + 3x^2$ $(d_1 = 2, d_2 = 3, d_3 = d_4 = 0)$

$e_1 = -2$

$e_2 = -(2 \times (-2) + 3) = 1$

$e_3 = -(2 \times 1 + 3 \times (-2)) = 4$

$e_4 = -(2 \times 4 + 3 \times 1) = -11 \quad \Rightarrow 1/q = 1 - 2x + x^2 + 4x^3 - 11x^4 + \ldots$

Why: the product $q \cdot (1/q)$, coefficient by coefficient

×	$e_0 = 1$	$e_1 = -2$	$e_2 = 1$	$e_3 = 4$	$e_4 = -11$
$d_0 = 1$	1	−2	1	4	−11
$d_1 = 2$	2	−4	2	8	−22
$d_2 = 3$	3	−6	3	12	−33

Each anti-diagonal ($i + j = k$) collects the coefficient of x^k. Highlighted, $k = 3$: $4 + 2 \times 1 + 3 \times (-2) = 0$ ✓

Every anti-diagonal sums to 0 (except $k = 0$, which gives 1). Solving each one for its newest entry e_k is exactly the rule in the yellow box.

Tool 2: clean-up by equivalence transformation

Multiplying the numerator and the denominator of a level by the same λ ≠ 0 leaves the value unchanged.

$a'_k = \lambda_{k-1} \cdot \lambda_k \cdot a_k\,, \quad b'_k = \lambda_k \cdot b_k\,, \quad \lambda_0 = 1$

Before (raw, from the algorithm)

$$\ln(1+x) = \cfrac{x}{1 + \cfrac{x/2}{1 + \cfrac{x/6}{1 + \cfrac{}{1 + \cdots}}}}$$

λ = 1, 2, 3 →

After (integer form)

$$\ln(1+x) = \cfrac{x}{1 + \cfrac{x}{2 + \cfrac{x}{3 + \cdots}}}$$

$a'_2 = \lambda_1\lambda_2 \cdot x/2 = 1 \times 2 \times x/2 = x,\ b'_2 = \lambda_2 = 2; \quad a'_3 = \lambda_2\lambda_3 \cdot x/6 = 2 \times 3 \times x/6 = x,\ b'_3 = \lambda_3 = 3$

How to choose λ_k: take the denominator still left at level k (λ_k = denominator of $\lambda_{k-1}\cdot\alpha_k$), or the value that makes b_k follow a visible pattern (k, 2k − 1, …).

Two ways to evaluate a truncated continued fraction

A Backward (bottom-up) scheme

$$t_n = b_n$$

$$t_k = b_k + a_{k+1} / t_{k+1}$$

$$k = n-1, n-2, \ldots, 1$$

$$f \approx b_0 + a_1 / t_1$$

✓ only n divisions, numerically stable
✗ n must be fixed before starting

B Forward recurrence (Wallis–Euler)

$$A_{-1} = 1,\ A_0 = b_0,\ \ B_{-1} = 0,\ B_0 = 1$$

$$A_k = b_k \cdot A_{k-1} + a_k \cdot A_{k-2}$$

$$B_k = b_k \cdot B_{k-1} + a_k \cdot B_{k-2}$$

$$C_k = A_k / B_k$$

✓ gives every convergent $C_1, C_2, \ldots$ in one sweep
✓ stop when $|C_k - C_{k-1}| < \varepsilon$

Why continued fractions? Same information, far better accuracy

Task: ln 2 = ln(1 + x) at x = 1, using the same 7 Taylor coefficients 1, −1/2, 1/3, …, 1/7

Taylor polynomial, degree 7

0.7595238095

0.0664

absolute error

Continued fraction, 7 levels

0.6931524548

5.27×10^{-6}

absolute error

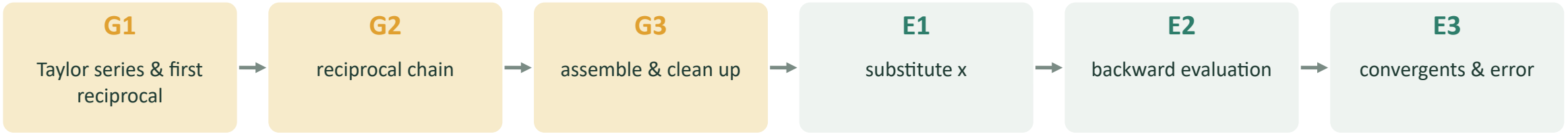
- The CF is built from exactly the same 7 coefficients, yet its error is about 12,600 times smaller.
- To match it, the Taylor series of ln 2 needs about 95 000 terms (its error behaves like 1/(2N)).
- A CF approximant is a rational function (Padé approximant): it follows poles and asymptotes that polynomials cannot.
- ln(1+x): the CF converges for every x > −1, the Taylor series only for −1 < x ≤ 1.

ln 2 = 0.6931471806
Taylor: 1 − 1/2 + 1/3 − 1/4 + 1/5 − 1/6 + 1/7
CF: 1/(1 + 1/(2 + 1/(3 + 4/(4 + 4/(5 + 9/(6 + 9/7))))))

The five examples: from starting point to generated fraction

#	Function	Start from	Generated continued fraction (integer form)
1	$\mathbf{e^x}$	$1 + x + x^2/2 + x^3/6 + \ldots$	$1 + x/(1 - x/(2 + x/(3 - x/(2 + x/(5 - \cdots)))))$
2	$\mathbf{\tan x}$	$x + x^3/3 + 2x^5/15 + \ldots$	$x/(1 - x^2/(3 - x^2/(5 - x^2/(7 - \cdots))))$
3	$\mathbf{\ln(1 + x)}$	$x - x^2/2 + x^3/3 - \ldots$	$x/(1 + x/(2 + x/(3 + 4x/(4 + 4x/(5 + \cdots)))))$
4	$\mathbf{\arctan x}$	$x - x^3/3 + x^5/5 - \ldots$	$x/(1 + x^2/(3 + 4x^2/(5 + 9x^2/(7 + \cdots))))$
5	$\mathbf{\sqrt{10}}$	$x^2 = 10$ (algebraic)	$3 + 1/(6 + 1/(6 + 1/(6 + \cdots)))$

Every example runs through the same six stages: G = generation (new), E = evaluation



CF Ex. 1 · e^x · G1: Taylor series and first reciprocal

Start $f_0 = e^x = 1 + x + x^2/2 + x^3/6 + x^4/24 + x^5/120 + x^6/720 + x^7/5040 + \dots$

1 **Constant term:** $\beta_0 = 1$

2 **Lowest term of the rest:** $f_0 - \beta_0 = x \cdot (1 + x/2 + x^2/6 + x^3/24 + x^4/120 + x^5/720 + x^6/5040 + \dots) \Rightarrow \alpha_1 x^m = x$

3 **Invert the bracket q with Tool 1:** $e_k = -(d_1 e_{k-1} + \dots + d_k)$

$q = 1 + x/2 + x^2/6 + x^3/24 + x^4/120 + x^5/720 + x^6/5040$

$d_1 = 1/2,\ d_2 = 1/6,\ d_3 = 1/24,\ d_4 = 1/120,\ d_5 = 1/720,\ d_6 = 1/5040$

$e_0 = 1$

$e_1 = -d_1 = -(1/2) = -1/2$

$e_2 = -(d_1 e_1 + d_2) = -((1/2)(-1/2) + (1/6)) = 1/12$

$e_3 = -(d_1 e_2 + d_2 e_1 + d_3) = -((1/2)(1/12) + (1/6)(-1/2) + (1/24)) = 0$

$e_4 = -(d_1 e_3 + d_2 e_2 + d_3 e_1 + d_4) = -((1/2)(0) + (1/6)(1/12) + (1/24)(-1/2) + (1/120)) = -1/720$

$e_5 \dots e_6$: same rule $\rightarrow$ $0, 1/30240$

$f_1 = 1/q = 1 - x/2 + x^2/12 - x^4/720 + x^6/30240 + \dots$ → repeat steps 1–3 on f_1 (G2)

CF Ex. 1 · e^x · G2: reciprocal chain (1 of 2)

0

$f_0 = 1 + x + x^2/2 + x^3/6 + x^4/24 + x^5/120 + x^6/720 + x^7/5040 + \ldots \Rightarrow \beta_0 = 1$

$f_0 - \beta_0 = x \cdot (1 + x/2 + x^2/6 + x^3/24 + x^4/120 + x^5/720 + x^6/5040 + \ldots)$ ⇒ next numerator **x**

$f_1 = 1 / (\text{bracket above}) = 1 - x/2 + x^2/12 - x^4/720 + x^6/30240 + \ldots$

1

$f_1 = 1 - x/2 + x^2/12 - x^4/720 + x^6/30240 + \ldots \Rightarrow \beta_1 = 1$

$f_1 - \beta_1 = (-x/2) \cdot (1 - x/6 + x^3/360 - x^5/15120 + \ldots)$ ⇒ next numerator **$-x/2$**

$f_2 = 1 / (\text{bracket above}) = 1 + x/6 + x^2/36 + x^3/540 - x^4/6480 - x^5/27216 + \ldots$

2

$f_2 = 1 + x/6 + x^2/36 + x^3/540 - x^4/6480 - x^5/27216 + \ldots \Rightarrow \beta_2 = 1$

$f_2 - \beta_2 = (x/6) \cdot (1 + x/6 + x^2/90 - x^3/1080 - x^4/4536 + \ldots)$ ⇒ next numerator **$x/6$**

$f_3 = 1 / (\text{bracket above}) = 1 - x/6 + x^2/60 - x^4/8400 + \ldots$

3

$f_3 = 1 - x/6 + x^2/60 - x^4/8400 + \ldots \Rightarrow \beta_3 = 1$

$f_3 - \beta_3 = (-x/6) \cdot (1 - x/10 + x^3/1400 + \ldots)$ ⇒ next numerator **$-x/6$**

$f_4 = 1 / (\text{bracket above}) = 1 + x/10 + x^2/100 + x^3/3500 + \ldots$

CF Ex. 1 · e^x · G2: reciprocal chain (2 of 2)

4

$f_4 = 1 + x/10 + x^2/100 + x^3/3500 + \ldots \Rightarrow \beta_4 = 1$

$f_4 - \beta_4 = (x/10) \cdot (1 + x/10 + x^2/350 + \ldots) \Rightarrow$ next numerator $x/10$

$f_5 = 1 / (\text{bracket above}) = 1 - x/10 + x^2/140 + \ldots$

5

$f_5 = 1 - x/10 + x^2/140 + \ldots \Rightarrow \beta_5 = 1$

$f_5 - \beta_5 = (-x/10) \cdot (1 - x/14 + \ldots) \Rightarrow$ next numerator $-x/10$

$f_6 = 1 / (\text{bracket above}) = 1 + x/14 + \ldots$

6

$f_6 = 1 + x/14 + \ldots \Rightarrow \beta_6 = 1$

$f_6 - \beta_6 = (x/14) \cdot (1 + \ldots) \Rightarrow$ next numerator $x/14$

last level: $f_7 \approx 1 \rightarrow$ stop

Collected: $\beta_0 = 1$, $\beta_k = 1$ $(k \geq 1)$; numerators $x, -x/2, x/6, -x/6, x/10, -x/10, x/14$

→ G3 stacks them into the continued fraction

CF Ex. 1 · e^x · G3: assemble and clean up

Raw: β_k as leads, numerators from G2

$$e^x = 1 + \cfrac{x}{1 - \cfrac{x/2}{1 + \cfrac{x/6}{1 - \cfrac{x/6}{1 + \cdots}}}}$$

Integer form after Tool 2 (λ from the table)

$$e^x = 1 + \cfrac{x}{1 - \cfrac{x}{2 + \cfrac{x}{3 - \cfrac{x}{2 + \cdots}}}}$$

k	$\alpha_k x^m$	λ_k	$a_k = \lambda_{k-1}\cdot\lambda_k\cdot\alpha_k x^m$	$b_k = \lambda_k$
1	x	1	1 × 1 × (x) = x	1
2	−x/2	2	1 × 2 × (−x/2) = −x	2
3	x/6	3	2 × 3 × (x/6) = x	3
4	−x/6	2	3 × 2 × (−x/6) = −x	2
5	x/10	5	2 × 5 × (x/10) = x	5
6	−x/10	2	5 × 2 × (−x/10) = −x	2
7	x/14	7	2 × 7 × (x/14) = x	7

Choice of λ: λ_k = denominator left at level k: 1, 2, 3, 2, 5, 2, 7

Pattern: $a_k = \pm x$ (alternating), $b_k = k$ (k odd), 2 (k even)

With the pattern known, the fraction can be continued to any depth without new series work.

CF Ex. 1 · e^x · E1: substitute x and tabulate

Goal: evaluate e = e^1 with n = 7 levels

$$e^x = 1 + \cfrac{x}{1 - \cfrac{x}{2 + \cfrac{x}{3 - \cfrac{x}{2 + \cdots}}}}$$

k	0	1	2	3	4	5	6	7
b_k	1	1	2	3	2	5	2	7
a_k	—	1	−1	1	−1	1	−1	1

1 Pattern of a_k and b_k (from G3)

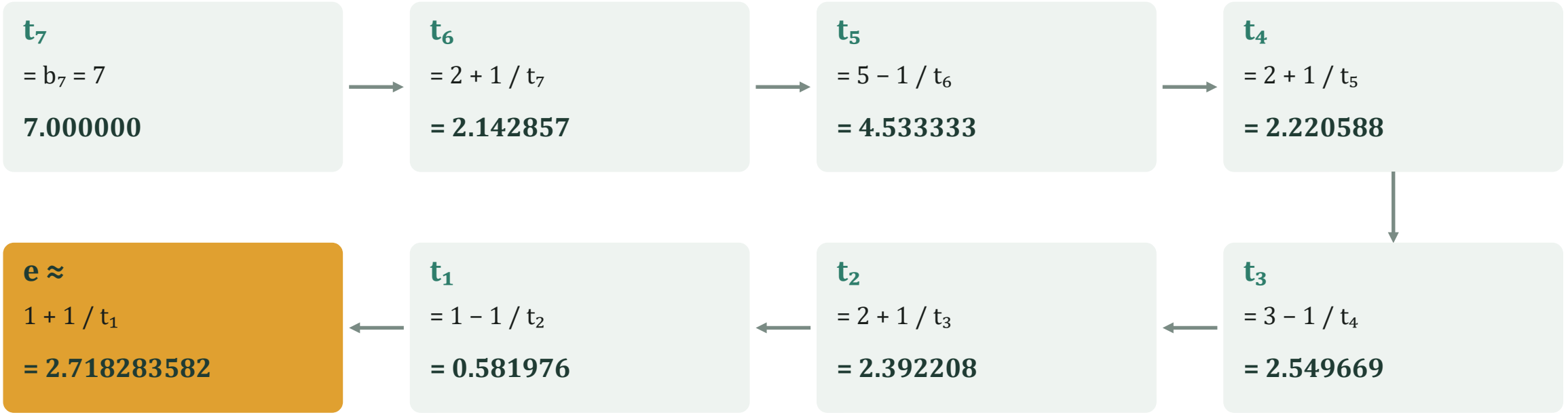
$b_0 = 1$, $a_k = x, -x, x, -x, \ldots$ (alternating)
$b_k = k$ for odd k, $b_k = 2$ for even k

2 Substitute the numerical data

$x = 1 \Rightarrow a_k = 1, -1, 1, -1, 1, -1, 1$; $b_k = 1, 2, 3, 2, 5, 2, 7$

Truncation: keep 7 levels, i.e. compute the convergent C_7. The table (left, bottom) is the complete input of both algorithms.

CF Ex. 1 · e^x · E2: backward evaluation (bottom-up)

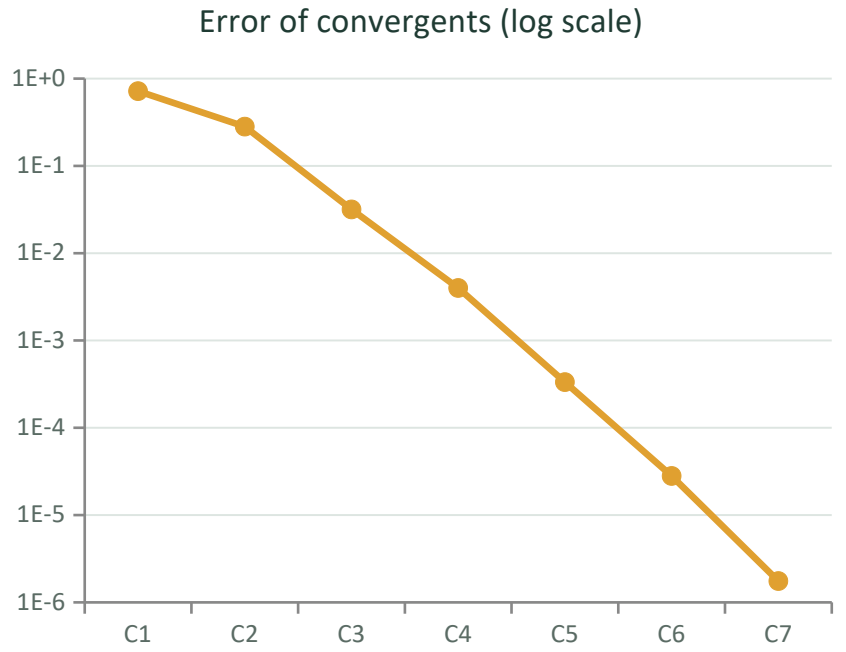


Result: $e \approx 2.718283582$ Exact: e = 2.718281828459…
Absolute error: 1.75×10^{-6} obtained with only 7 divisions. Read the snake in the arrow direction: start at the deepest level t_7.

CF Ex. 1 · e^x · E3: convergents by forward recurrence

$A_k = b_k \cdot A_{k-1} + a_k \cdot A_{k-2}$ $B_k = b_k \cdot B_{k-1} + a_k \cdot B_{k-2}$ $C_k = A_k / B_k$ $(A_{-1} = 1, A_0 = b_0, B_{-1} = 0, B_0 = 1)$

k	a_k	b_k	A_k	B_k	$C_k = A_k/B_k$	$\lvert C_k - f \rvert$
0	—	1	1	1	1.000000000	1.7183
1	1	1	2	1	2.000000000	0.7183
2	−1	2	3	1	3.000000000	0.2817
3	1	3	11	4	2.750000000	0.0317
4	−1	2	19	7	2.714285714	4.00×10^{-3}
5	1	5	106	39	2.717948718	3.33×10^{-4}
6	−1	2	193	71	2.718309859	2.80×10^{-5}
7	1	7	1457	536	2.718283582	1.75×10^{-6}



$C_7 = 2.718283582 \equiv$ backward result ✓
≈ 5 correct decimals

CF Ex. 2 · tan x · G1: Taylor series and first reciprocal

Start $f_0 = \tan x = x + x^3/3 + 2x^5/15 + 17x^7/315 + 62x^9/2835 + \ldots$ (series of tan x from sin x · (1/cos x), Tool 1)

1 **Constant term:** $\beta_0 = 0$

2 **Lowest term of the rest:** $f_0 - \beta_0 = x \cdot (1 + x^2/3 + 2x^4/15 + 17x^6/315 + 62x^8/2835 + \ldots) \Rightarrow \alpha_1 x^m = x$

3 **Invert the bracket q with Tool 1:** $e_k = -(d_1 e_{k-1} + \ldots + d_k)$

only even powers → put $t = x^2$: $q = 1 + t/3 + 2t^2/15 + 17t^3/315 + 62t^4/2835$

$d_1 = 1/3, d_2 = 2/15, d_3 = 17/315, d_4 = 62/2835$

$e_0 = 1$

$e_1 = -d_1 = -(1/3) = -1/3$

$e_2 = -(d_1 e_1 + d_2) = -((1/3)(-1/3) + (2/15)) = -1/45$

$e_3 = -(d_1 e_2 + d_2 e_1 + d_3) = -((1/3)(-1/45) + (2/15)(-1/3) + (17/315)) = -2/945$

$e_4 = -(d_1 e_3 + d_2 e_2 + d_3 e_1 + d_4) = -((1/3)(-2/945) + (2/15)(-1/45) + (17/315)(-1/3) + (62/2835)) = -1/4725$

$f_1 = 1/q = 1 - x^2/3 - x^4/45 - 2x^6/945 - x^8/4725 + \ldots$ → repeat steps 1–3 on f_1 (G2)

CF Ex. 2 · tan x · G2: reciprocal chain (1 of 2)

0

$f_0 = x + x^3/3 + 2x^5/15 + 17x^7/315 + 62x^9/2835 + \ldots \Rightarrow \beta_0 = 0$

$f_0 - \beta_0 = x \cdot (1 + x^2/3 + 2x^4/15 + 17x^6/315 + 62x^8/2835 + \ldots) \Rightarrow$ next numerator x

$f_1 = 1 / (\text{bracket above}) = 1 - x^2/3 - x^4/45 - 2x^6/945 - x^8/4725 + \ldots$

1

$f_1 = 1 - x^2/3 - x^4/45 - 2x^6/945 - x^8/4725 + \ldots \Rightarrow \beta_1 = 1$

$f_1 - \beta_1 = (-x^2/3) \cdot (1 + x^2/15 + 2x^4/315 + x^6/1575 + \ldots) \Rightarrow$ next numerator $-x^2/3$

$f_2 = 1 / (\text{bracket above}) = 1 - x^2/15 - x^4/525 - 2x^6/23625 + \ldots$

2

$f_2 = 1 - x^2/15 - x^4/525 - 2x^6/23625 + \ldots \Rightarrow \beta_2 = 1$

$f_2 - \beta_2 = (-x^2/15) \cdot (1 + x^2/35 + 2x^4/1575 + \ldots) \Rightarrow$ next numerator $-x^2/15$

$f_3 = 1 / (\text{bracket above}) = 1 - x^2/35 - x^4/2205 + \ldots$

CF Ex. 2 · tan x · G2: reciprocal chain (2 of 2)

3

$f_3 = 1 - x^2/35 - x^4/2205 + \ldots \Rightarrow \beta_3 = 1$

$f_3 - \beta_3 = (-x^2/35) \cdot (1 + x^2/63 + \ldots) \Rightarrow$ next numerator $-x^2/35$

$f_4 = 1 /$ (bracket above) $= 1 - x^2/63 + \ldots$

4

$f_4 = 1 - x^2/63 + \ldots \Rightarrow \beta_4 = 1$

$f_4 - \beta_4 = (-x^2/63) \cdot (1 + \ldots) \Rightarrow$ next numerator $-x^2/63$

last level: $f_5 \approx 1 \rightarrow$ stop

Collected: $\beta_0 = 0$, $\beta_k = 1$ $(k \geq 1)$; numerators $x, -x^2/3, -x^2/15, -x^2/35, -x^2/63$

→ G3 stacks them into the continued fraction

CF Ex. 2 · tan x · G3: assemble and clean up

Raw: β_k as leads, numerators from G2

$$\tan x = \cfrac{x}{1 - \cfrac{x^2/3}{1 - \cfrac{x^2/15}{1 - \cfrac{x^2/35}{1 - \cdots}}}}$$

Integer form after Tool 2 (λ from the table)

$$\tan x = \cfrac{x}{1 - \cfrac{x^2}{3 - \cfrac{x^2}{5 - \cfrac{x^2}{7 - \cdots}}}}$$

k	$\alpha_k x^m$	λ_k	$a_k = \lambda_{k-1}\cdot\lambda_k\cdot\alpha_k x^m$	$b_k = \lambda_k$
1	x	1	$1 \times 1 \times (x) = x$	1
2	$-x^2/3$	3	$1 \times 3 \times (-x^2/3) = -x^2$	3
3	$-x^2/15$	5	$3 \times 5 \times (-x^2/15) = -x^2$	5
4	$-x^2/35$	7	$5 \times 7 \times (-x^2/35) = -x^2$	7
5	$-x^2/63$	9	$7 \times 9 \times (-x^2/63) = -x^2$	9

Choice of λ: $\lambda_k = 2k - 1$ clears $\alpha_k = -1/((2k-3)(2k-1))$

Pattern: $a_1 = x$, $a_k = -x^2$ $(k \geq 2)$, $b_k = 2k - 1$

With the pattern known, the fraction can be continued to any depth without new series work.

CF Ex. 2 · tan x · E1: substitute x and tabulate

Goal: evaluate tan 1 with n = 5 levels

$$\tan x = \cfrac{x}{1 - \cfrac{x^2}{3 - \cfrac{x^2}{5 - \cfrac{x^2}{7 - \cdots}}}}$$

k	0	1	2	3	4	5
b_k	0	1	3	5	7	9
a_k	—	1	−1	−1	−1	−1

1 **Pattern of a_k and b_k (from G3)**

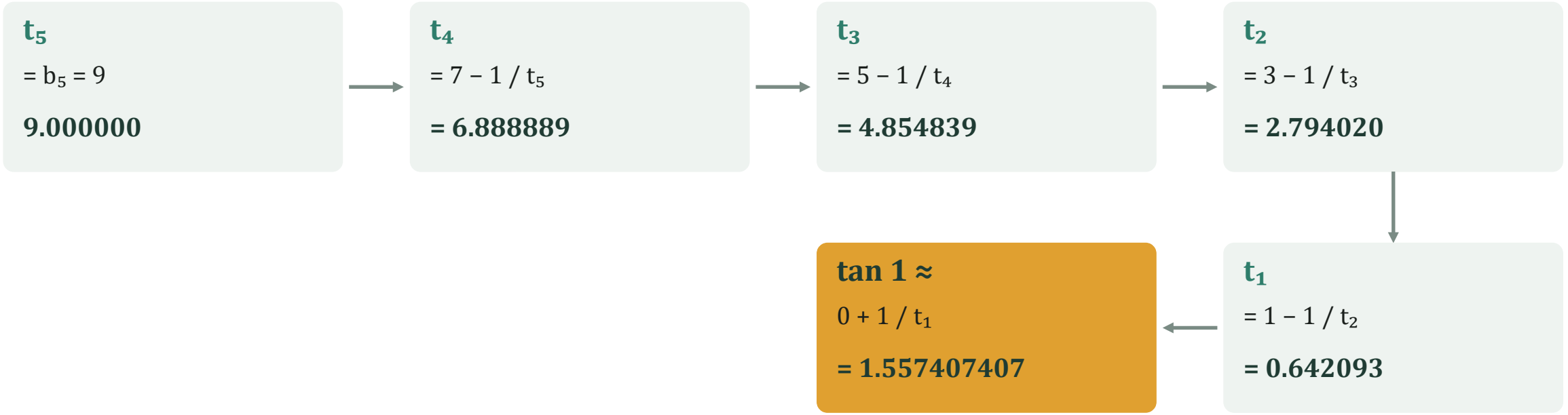
$b_0 = 0$, $a_1 = x$, $a_k = -x^2$ $(k \geq 2)$
$b_k = 2k - 1$ (odd numbers)

2 **Substitute the numerical data**

$x = 1$ (rad) $\Rightarrow$ $a_1 = 1$, $a_k = -1$, $b_k = 1, 3, 5, 7, 9$

Truncation: keep 5 levels, i.e. compute the convergent C_5. The table (left, bottom) is the complete input of both algorithms.

CF Ex. 2 · tan x · E2: backward evaluation (bottom-up)

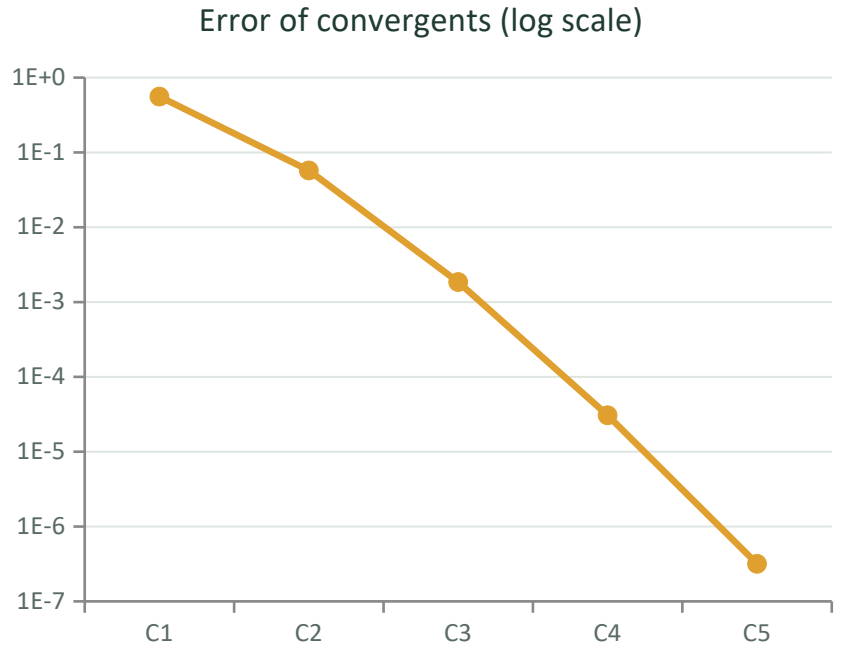


Result: **tan 1 ≈ 1.557407407** Exact: tan 1 = 1.557407724654…
Absolute error: 3.17×10^{-7} obtained with only 5 divisions. Read the snake in the arrow direction: start at the deepest level t_5.

CF Ex. 2 · tan x · E3: convergents by forward recurrence

$A_k = b_k \cdot A_{k-1} + a_k \cdot A_{k-2}$ $B_k = b_k \cdot B_{k-1} + a_k \cdot B_{k-2}$ $C_k = A_k / B_k$ $(A_{-1} = 1, A_0 = b_0, B_{-1} = 0, B_0 = 1)$

k	a_k	b_k	A_k	B_k	$C_k = A_k/B_k$	$\lvert C_k - f \rvert$
0	—	0	0	1	0.000000000	1.5574
1	1	1	1	1	1.000000000	0.5574
2	−1	3	3	2	1.500000000	0.0574
3	−1	5	14	9	1.555555556	1.85×10^{-3}
4	−1	7	95	61	1.557377049	3.07×10^{-5}
5	−1	9	841	540	1.557407407	3.17×10^{-7}



$C_5 = 1.557407407 \equiv$ backward result ✓
≈ 6 correct decimals

CF Ex. 3 · ln(1+x) · G1: Taylor series and first reciprocal

Start $f_0 = \ln(1+x) = x - x^2/2 + x^3/3 - x^4/4 + x^5/5 - x^6/6 + x^7/7 + \ldots$

1 **Constant term:** $\beta_0 = 0$

2 **Lowest term of the rest:** $f_0 - \beta_0 = x \cdot (1 - x/2 + x^2/3 - x^3/4 + x^4/5 - x^5/6 + x^6/7 + \ldots) \Rightarrow \alpha_1 x^m = x$

3 **Invert the bracket q with Tool 1:** $e_k = -(d_1 e_{k-1} + \ldots + d_k)$

$q = 1 - x/2 + x^2/3 - x^3/4 + x^4/5 - x^5/6 + x^6/7$

$d_1 = -1/2,\ d_2 = 1/3,\ d_3 = -1/4,\ d_4 = 1/5,\ d_5 = -1/6,\ d_6 = 1/7$

$e_0 = 1$

$e_1 = -d_1 = -(-1/2) = 1/2$

$e_2 = -(d_1 e_1 + d_2) = -((-1/2)(1/2) + (1/3)) = -1/12$

$e_3 = -(d_1 e_2 + d_2 e_1 + d_3) = -((-1/2)(-1/12) + (1/3)(1/2) + (-1/4)) = 1/24$

$e_4 = -(d_1 e_3 + d_2 e_2 + d_3 e_1 + d_4) = -((-1/2)(1/24) + (1/3)(-1/12) + (-1/4)(1/2) + (1/5)) = -19/720$

$e_5 \ldots e_6$: same rule → $3/160, -863/60480$

$f_1 = 1/q = 1 + x/2 - x^2/12 + x^3/24 - 19x^4/720 + 3x^5/160 - 863x^6/60480 + \ldots$ → repeat steps 1–3 on f_1 (G2)

CF Ex. 3 · ln(1+x) · G2: reciprocal chain (1 of 2)

0

$f_0 = x - x^2/2 + x^3/3 - x^4/4 + x^5/5 - x^6/6 + x^7/7 + \ldots \Rightarrow \beta_0 = 0$

$f_0 - \beta_0 = x \cdot (1 - x/2 + x^2/3 - x^3/4 + x^4/5 - x^5/6 + x^6/7 + \ldots) \Rightarrow$ next numerator x

$f_1 = 1 / (\text{bracket above}) = 1 + x/2 - x^2/12 + x^3/24 - 19x^4/720 + 3x^5/160 - 863x^6/60480 + \ldots$

1

$f_1 = 1 + x/2 - x^2/12 + x^3/24 - 19x^4/720 + 3x^5/160 - 863x^6/60480 + \ldots \Rightarrow \beta_1 = 1$

$f_1 - \beta_1 = (x/2) \cdot (1 - x/6 + x^2/12 - 19x^3/360 + 3x^4/80 - 863x^5/30240 + \ldots) \Rightarrow$ next numerator $x/2$

$f_2 = 1 / (\text{bracket above}) = 1 + x/6 - x^2/18 + 4x^3/135 - 31x^4/1620 + 233x^5/17010 + \ldots$

2

$f_2 = 1 + x/6 - x^2/18 + 4x^3/135 - 31x^4/1620 + 233x^5/17010 + \ldots \Rightarrow \beta_2 = 1$

$f_2 - \beta_2 = (x/6) \cdot (1 - x/3 + 8x^2/45 - 31x^3/270 + 233x^4/2835 + \ldots) \Rightarrow$ next numerator $x/6$

$f_3 = 1 / (\text{bracket above}) = 1 + x/3 - x^2/15 + x^3/30 - 11x^4/525 + \ldots$

3

$f_3 = 1 + x/3 - x^2/15 + x^3/30 - 11x^4/525 + \ldots \Rightarrow \beta_3 = 1$

$f_3 - \beta_3 = (x/3) \cdot (1 - x/5 + x^2/10 - 11x^3/175 + \ldots) \Rightarrow$ next numerator $x/3$

$f_4 = 1 / (\text{bracket above}) = 1 + x/5 - 3x^2/50 + 27x^3/875 + \ldots$

CF Ex. 3 · ln(1+x) · G2: reciprocal chain (2 of 2)

4

$f_4 = 1 + x/5 - 3x^2/50 + 27x^3/875 + \ldots \Rightarrow \beta_4 = 1$

$f_4 - \beta_4 = (x/5) \cdot (1 - 3x/10 + 27x^2/175 + \ldots) \Rightarrow$ next numerator $x/5$

$f_5 = 1 / (\text{bracket above}) = 1 + 3x/10 - 9x^2/140 + \ldots$

5

$f_5 = 1 + 3x/10 - 9x^2/140 + \ldots \Rightarrow \beta_5 = 1$

$f_5 - \beta_5 = (3x/10) \cdot (1 - 3x/14 + \ldots) \Rightarrow$ next numerator $3x/10$

$f_6 = 1 / (\text{bracket above}) = 1 + 3x/14 + \ldots$

6

$f_6 = 1 + 3x/14 + \ldots \Rightarrow \beta_6 = 1$

$f_6 - \beta_6 = (3x/14) \cdot (1 + \ldots) \Rightarrow$ next numerator $3x/14$

last level: $f_7 \approx 1 \rightarrow$ stop

Collected: $\beta_0 = 0$, $\beta_k = 1$ $(k \geq 1)$; numerators $x,\ x/2,\ x/6,\ x/3,\ x/5,\ 3x/10,\ 3x/14$

→ G3 stacks them into the continued fraction

CF Ex. 3 · ln(1+x) · G3: assemble and clean up

Raw: β_k as leads, numerators from G2

$$\ln(1+x) = \cfrac{x}{1+\cfrac{x/2}{1+\cfrac{x/6}{1+\cfrac{x/3}{1+\cdots}}}}$$

Integer form after Tool 2 (λ from the table)

$$\ln(1+x) = \cfrac{x}{1+\cfrac{x}{2+\cfrac{x}{3+\cfrac{4x}{4+\cdots}}}}$$

k	$\alpha_k x^m$	λ_k	$a_k = \lambda_{k-1}\cdot\lambda_k\cdot\alpha_k x^m$	$b_k = \lambda_k$
1	x	1	1 × 1 × (x) = x	1
2	x/2	2	1 × 2 × (x/2) = x	2
3	x/6	3	2 × 3 × (x/6) = x	3
4	x/3	4	3 × 4 × (x/3) = 4x	4
5	x/5	5	4 × 5 × (x/5) = 4x	5
6	3x/10	6	5 × 6 × (3x/10) = 9x	6
7	3x/14	7	6 × 7 × (3x/14) = 9x	7

Choice of λ: $\lambda_k = k$

Pattern: a_k = x, x, x, 4x, 4x, 9x, 9x (each m^2x twice), $b_k = k$

With the pattern known, the fraction can be continued to any depth without new series work.

CF Ex. 3 · ln(1+x) · E1: substitute x and tabulate

Goal: evaluate ln 2 with n = 7 levels

$$\ln(1+x) = \cfrac{x}{1+\cfrac{x}{2+\cfrac{x}{3+\cfrac{4x}{4+\cdots}}}}$$

k	0	1	2	3	4	5	6	7
b_k	0	1	2	3	4	5	6	7
a_k	—	1	1	1	4	4	9	9

1 **Pattern of a_k and b_k (from G3)**

$b_0 = 0,\ b_k = k,\ a_1 = x$

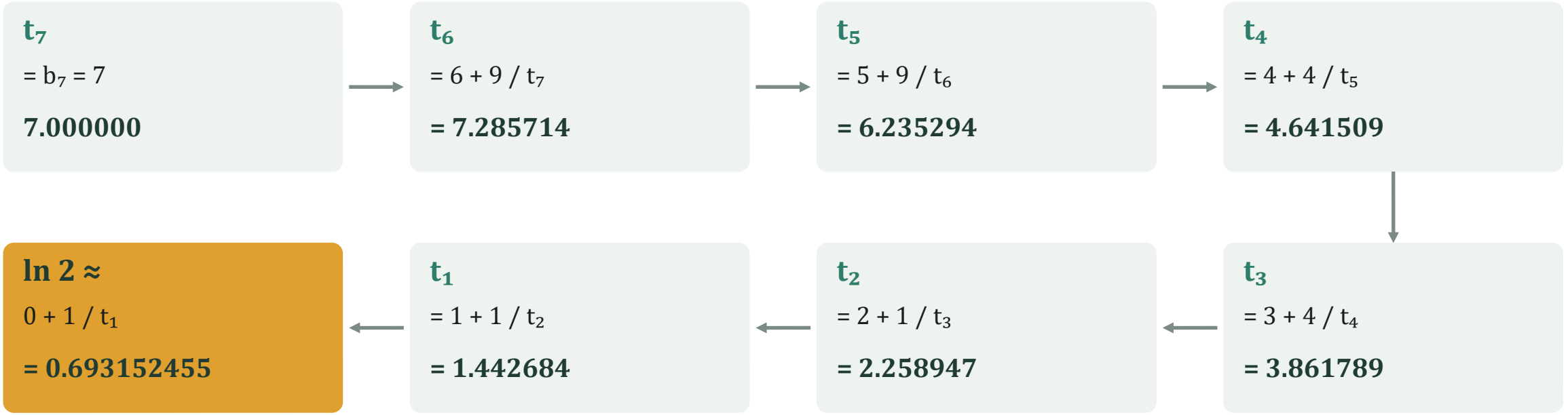
$a_{2m} = a_{2m+1} = m^2 \cdot x \quad (m \geq 1)$

2 **Substitute the numerical data**

$x = 1 \Rightarrow a_k = 1, 1, 1, 4, 4, 9, 9;\ b_k = 1, 2, \ldots, 7$

Truncation: keep 7 levels, i.e. compute the convergent C_7. The table (left, bottom) is the complete input of both algorithms.

CF Ex. 3 · ln(1+x) · E2: backward evaluation (bottom-up)

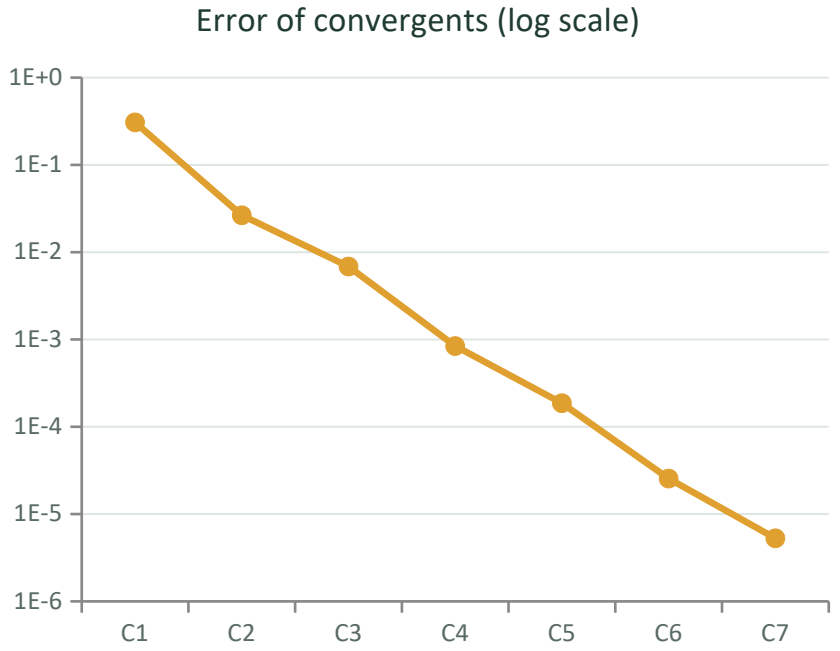


Result: $\ln 2 \approx 0.693152455$ Exact: $\ln 2 = 0.693147180559\ldots$
Absolute error: 5.27×10^{-6} obtained with only 7 divisions. Read the snake in the arrow direction: start at the deepest level t_7.

CF Ex. 3 · ln(1+x) · E3: convergents by forward recurrence

$A_k = b_k \cdot A_{k-1} + a_k \cdot A_{k-2}$ $B_k = b_k \cdot B_{k-1} + a_k \cdot B_{k-2}$ $C_k = A_k / B_k$ $(A_{-1} = 1, A_0 = b_0, B_{-1} = 0, B_0 = 1)$

k	a_k	b_k	A_k	B_k	$C_k = A_k/B_k$	$\lvert C_k - f \rvert$
0	—	0	0	1	0.000000000	0.6931
1	1	1	1	1	1.000000000	0.3069
2	1	2	2	3	0.666666667	0.0265
3	1	3	7	10	0.700000000	6.85×10^{-3}
4	4	4	36	52	0.692307692	8.39×10^{-4}
5	4	5	208	300	0.693333333	1.86×10^{-4}
6	9	6	1572	2268	0.693121693	2.55×10^{-5}
7	9	7	12876	18576	0.693152455	5.27×10^{-6}



$C_7 = 0.693152455 \equiv$ backward result ✓
≈ 5 correct decimals

CF Ex. 4 · arctan x · G1: Taylor series and first reciprocal

Start $f_0 = \arctan x = x - x^3/3 + x^5/5 - x^7/7 + x^9/9 - x^{11}/11 + \ldots$

1 **Constant term:** $\beta_0 = 0$

2 **Lowest term of the rest:** $f_0 - \beta_0 = x \cdot (1 - x^2/3 + x^4/5 - x^6/7 + x^8/9 - x^{10}/11 + \ldots) \Rightarrow \alpha_1 x^m = x$

3 **Invert the bracket q with Tool 1:** $e_k = -(d_1 e_{k-1} + \ldots + d_k)$

only even powers → put $t = x^2$: $q = 1 - t/3 + t^2/5 - t^3/7 + t^4/9 - t^5/11$

$d_1 = -1/3,\ d_2 = 1/5,\ d_3 = -1/7,\ d_4 = 1/9,\ d_5 = -1/11$

$e_0 = 1$

$e_1 = -d_1 = -(-1/3) = 1/3$

$e_2 = -(d_1 e_1 + d_2) = -((-1/3)(1/3) + (1/5)) = -4/45$

$e_3 = -(d_1 e_2 + d_2 e_1 + d_3) = -((-1/3)(-4/45) + (1/5)(1/3) + (-1/7)) = 44/945$

$e_4 = -(d_1 e_3 + d_2 e_2 + d_3 e_1 + d_4) = -((-1/3)(44/945) + (1/5)(-4/45) + (-1/7)(1/3) + (1/9)) = -428/14175$

e_5: same rule → $10196/467775$

$f_1 = 1/q = 1 + x^2/3 - 4x^4/45 + 44x^6/945 - 428x^8/14175 + 10196x^{10}/467775 + \ldots$ → repeat steps 1–3 on f_1 (G2)

CF Ex. 4 · arctan x · G2: reciprocal chain (1 of 2)

0

$f_0 = x - x^3/3 + x^5/5 - x^7/7 + x^9/9 - x^{11}/11 + \ldots \Rightarrow \beta_0 = 0$

$f_0 - \beta_0 = x \cdot (1 - x^2/3 + x^4/5 - x^6/7 + x^8/9 - x^{10}/11 + \ldots) \Rightarrow$ next numerator x

$f_1 = 1 / (\text{bracket above}) = 1 + x^2/3 - 4x^4/45 + 44x^6/945 - 428x^8/14175 + 10196x^{10}/467775 + \ldots$

1

$f_1 = 1 + x^2/3 - 4x^4/45 + 44x^6/945 - 428x^8/14175 + 10196x^{10}/467775 + \ldots \Rightarrow \beta_1 = 1$

$f_1 - \beta_1 = (x^2/3) \cdot (1 - 4x^2/15 + 44x^4/315 - 428x^6/4725 + 10196x^8/155925 + \ldots) \Rightarrow$ next numerator $x^2/3$

$f_2 = 1 / (\text{bracket above}) = 1 + 4x^2/15 - 12x^4/175 + 92x^6/2625 - 7516x^8/336875 + \ldots$

2

$f_2 = 1 + 4x^2/15 - 12x^4/175 + 92x^6/2625 - 7516x^8/336875 + \ldots \Rightarrow \beta_2 = 1$

$f_2 - \beta_2 = (4x^2/15) \cdot (1 - 9x^2/35 + 23x^4/175 - 5637x^6/67375 + \ldots) \Rightarrow$ next numerator $4x^2/15$

$f_3 = 1 / (\text{bracket above}) = 1 + 9x^2/35 - 16x^4/245 + 624x^6/18865 + \ldots$

CF Ex. 4 · arctan x · G2: reciprocal chain (2 of 2)

3

$f_3 = 1 + 9x^2/35 - 16x^4/245 + 624x^6/18865 + \ldots \Rightarrow \beta_3 = 1$

$f_3 - \beta_3 = (9x^2/35) \cdot (1 - 16x^2/63 + 208x^4/1617 + \ldots) \Rightarrow$ next numerator $9x^2/35$

$f_4 = 1 /$ (bracket above) $= 1 + 16x^2/63 - 400x^4/6237 + \ldots$

4

$f_4 = 1 + 16x^2/63 - 400x^4/6237 + \ldots \Rightarrow \beta_4 = 1$

$f_4 - \beta_4 = (16x^2/63) \cdot (1 - 25x^2/99 + \ldots) \Rightarrow$ next numerator $16x^2/63$

$f_5 = 1 /$ (bracket above) $= 1 + 25x^2/99 + \ldots$

5

$f_5 = 1 + 25x^2/99 + \ldots \Rightarrow \beta_5 = 1$

$f_5 - \beta_5 = (25x^2/99) \cdot (1 + \ldots) \Rightarrow$ next numerator $25x^2/99$

last level: $f_6 \approx 1 \rightarrow$ stop

Collected: $\beta_0 = 0$, $\beta_k = 1$ $(k \geq 1)$; numerators $x,\ x^2/3,\ 4x^2/15,\ 9x^2/35,\ 16x^2/63,\ 25x^2/99$

→ G3 stacks them into the continued fraction

CF Ex. 4 · arctan x · G3: assemble and clean up

Raw: β_k as leads, numerators from G2

$$\arctan x = \cfrac{x}{1 + \cfrac{x^2/3}{1 + \cfrac{4x^2/15}{1 + \cfrac{9x^2/35}{1 + \cfrac{}{1 + \cdots}}}}}$$

k	$\alpha_k x^m$	λ_k	$a_k = \lambda_{k-1}\cdot\lambda_k\cdot\alpha_k x^m$	$b_k = \lambda_k$
1	x	1	$1 \times 1 \times (x) = x$	1
2	$x^2/3$	3	$1 \times 3 \times (x^2/3) = x^2$	3
3	$4x^2/15$	5	$3 \times 5 \times (4x^2/15) = 4x^2$	5
4	$9x^2/35$	7	$5 \times 7 \times (9x^2/35) = 9x^2$	7
5	$16x^2/63$	9	$7 \times 9 \times (16x^2/63) = 16x^2$	9
6	$25x^2/99$	11	$9 \times 11 \times (25x^2/99) = 25x^2$	11

Integer form after Tool 2 (λ from the table)

$$\arctan x = \cfrac{x}{1 + \cfrac{x^2}{3 + \cfrac{4x^2}{5 + \cfrac{9x^2}{7 + \cdots}}}}$$

Choice of λ: $\lambda_k = 2k - 1$

Pattern: $a_1 = x,\ a_k = (k-1)^2x^2\ (k \geq 2),\ b_k = 2k - 1$

With the pattern known, the fraction can be continued to any depth without new series work.

CF Ex. 4 · arctan x · E1: substitute x and tabulate

Goal: evaluate arctan 1 = π/4 with n = 6 levels

$$\arctan x = \cfrac{x}{1 + \cfrac{x^2}{3 + \cfrac{4x^2}{5 + \cfrac{9x^2}{7 + \cdots}}}}$$

1 Pattern of a_k and b_k (from G3)

$b_0 = 0,\ a_1 = x,\ b_k = 2k - 1$
$a_k = (k-1)^2 \cdot x^2 \quad (k \geq 2)$

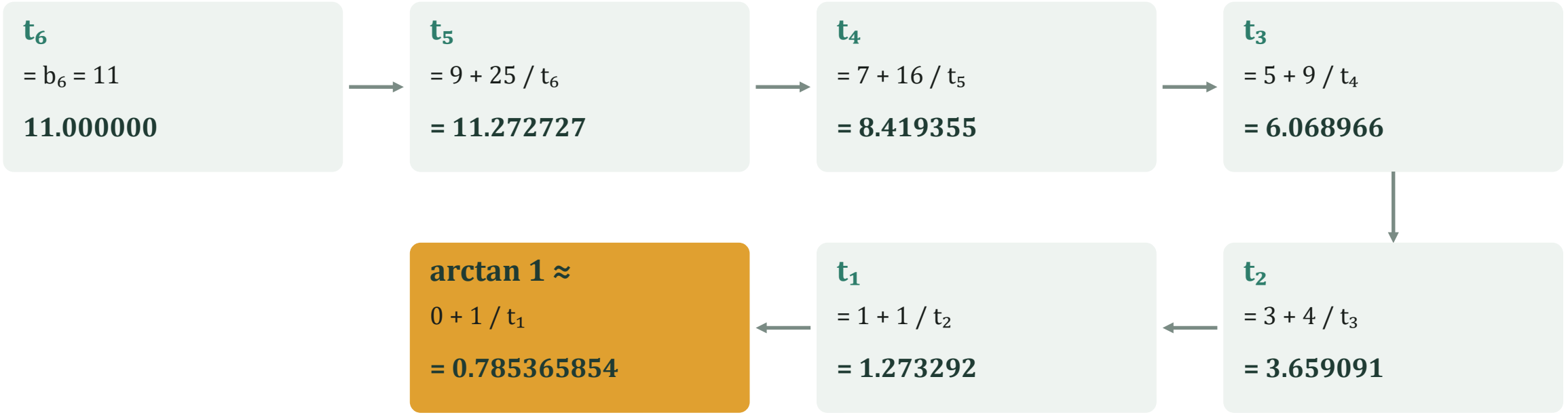
2 Substitute the numerical data

$x = 1 \Rightarrow a_k = 1, 1, 4, 9, 16, 25;\ b_k = 1, 3, \ldots, 11$

k	0	1	2	3	4	5	6
b_k	0	1	3	5	7	9	11
a_k	—	1	1	4	9	16	25

Truncation: keep 6 levels, i.e. compute the convergent C_6. The table (left, bottom) is the complete input of both algorithms.

CF Ex. 4 · arctan x · E2: backward evaluation (bottom-up)

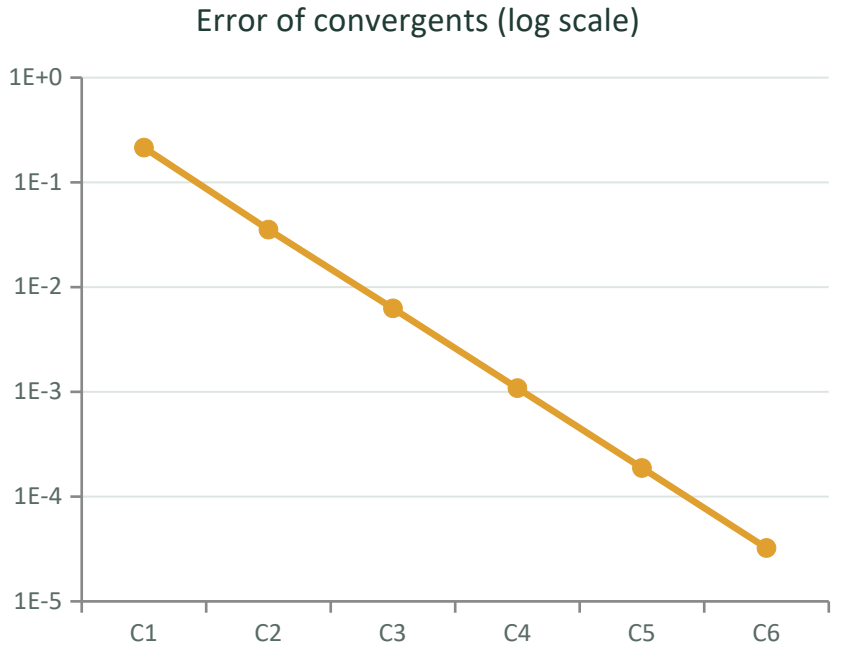


Result: arctan 1 ≈ 0.785365854 Exact: $\pi/4 = 0.785398163397\ldots$
Absolute error: 3.23×10^{-5} obtained with only 6 divisions. Read the snake in the arrow direction: start at the deepest level t_6.

CF Ex. 4 · arctan x · E3: convergents by forward recurrence

$A_k = b_k \cdot A_{k-1} + a_k \cdot A_{k-2}$ $B_k = b_k \cdot B_{k-1} + a_k \cdot B_{k-2}$ $C_k = A_k / B_k$ $(A_{-1} = 1, A_0 = b_0, B_{-1} = 0, B_0 = 1)$

k	a_k	b_k	A_k	B_k	$C_k = A_k/B_k$	$\lvert C_k - f \rvert$
0	—	0	0	1	0.000000000	0.7854
1	1	1	1	1	1.000000000	0.2146
2	1	3	3	4	0.750000000	0.0354
3	4	5	19	24	0.791666667	6.27×10^{-3}
4	9	7	160	204	0.784313725	1.08×10^{-3}
5	16	9	1744	2220	0.785585586	1.87×10^{-4}
6	25	11	23184	29520	0.785365854	3.23×10^{-5}



$C_6 = 0.785365854 \equiv$ backward result ✓
≈ 4 correct decimals

CF Ex. 5 · $\sqrt{10}$ · G1: generation by self-substitution

1 $x = \sqrt{10} \Rightarrow x^2 = 10 \Rightarrow x^2 - 3^2 = 1$

$3^2 = 9$ is the largest square below 10

2 $(x - 3)(x + 3) = 1 \Rightarrow x = 3 + 1/(3 + x)$

difference of squares, divide by (x + 3)

3 $x = 3 + 1/(3 + 3 + 1/(3 + x)) = 3 + 1/(6 + 1/(3 + x))$

replace x in the denominator by the whole right side

4 $x = 3 + 1/(6 + 1/(6 + 1/(3 + x)))$

and again: one more level each time

5 **tail 3 + x never disappears ⇒ infinite, periodic CF**

cut after n levels: replace 3 + x by 6

Result

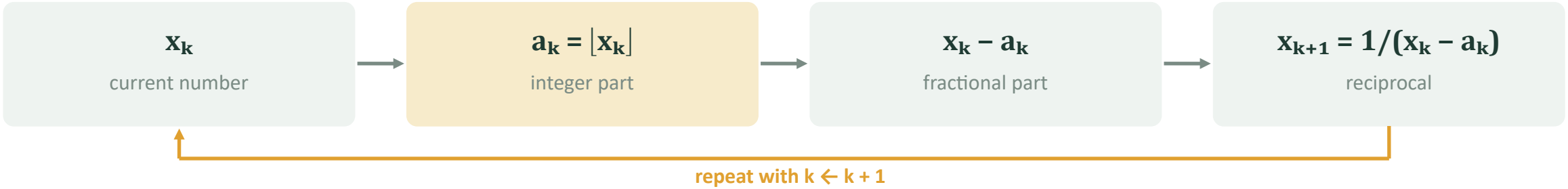
$$\sqrt{10} = 3 + \cfrac{1}{6 + \cfrac{1}{6 + \cfrac{1}{6 + \cdots}}}$$

General recipe

$x^2 = a^2 + b \Rightarrow (x - a)(x + a) = b$

$\sqrt{(a^2 + b)} = a + b/(2a + b/(2a + \cdots))$

CF Ex. 5 · $\sqrt{10}$ · G2: check with the integer-part algorithm



k	x_k (exact)	x_k (decimal)	a_k	$x_{k+1} = 1/(x_k - a_k)$ (rationalise)
0	$\sqrt{10}$	3.162278	3	$1/(\sqrt{10} - 3) = (\sqrt{10} + 3)/(10 - 9) = \sqrt{10} + 3$
1	$\sqrt{10} + 3$	6.162278	6	$1/(\sqrt{10} + 3 - 6) = 1/(\sqrt{10} - 3) = \sqrt{10} + 3$
2	$\sqrt{10} + 3$	6.162278	6	same as row 1 → period found

Rationalising trick:
$1/(\sqrt{10} - 3) \cdot (\sqrt{10} + 3)/(\sqrt{10} + 3) = (\sqrt{10} + 3)/1$

$\sqrt{10} = [3; 6, 6, 6, \ldots] = 3 + 1/(6 + 1/(6 + \cdots))$
same fraction as G1. This algorithm works for any real number.

CF Ex. 5 · √10 · E1: tabulate the coefficients

Goal: evaluate √10 with n = 5 levels

$$\sqrt{10} = 3 + \cfrac{1}{6 + \cfrac{1}{6 + \cfrac{1}{6 + \cdots}}}$$

k	0	1	2	3	4	5
b_k	3	6	6	6	6	6
a_k	—	1	1	1	1	1

1 **Pattern of a_k and b_k (from G3)**

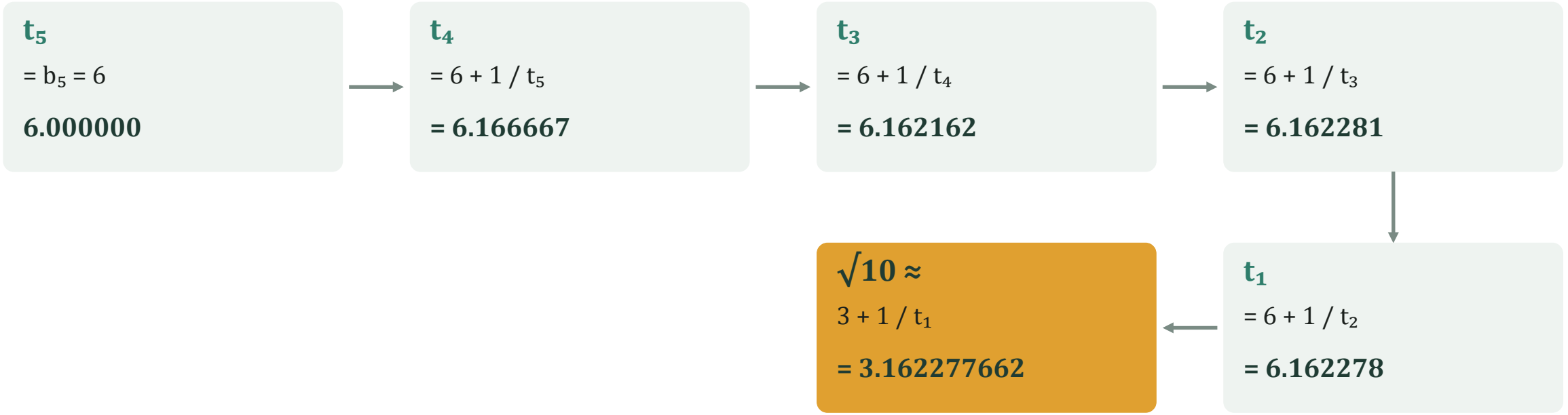
$b_0 = 3,\ a_k = 1,\ b_k = 6$ (periodic)
from $x = 3 + 1/(3 + x)$ (stage G1)

2 **Numbers only**

no variable to substitute: all coefficients are numbers

Truncation: keep 5 levels, i.e. compute the convergent C_5. The table (left, bottom) is the complete input of both algorithms.

CF Ex. 5 · $\sqrt{10}$ · E2: backward evaluation (bottom-up)

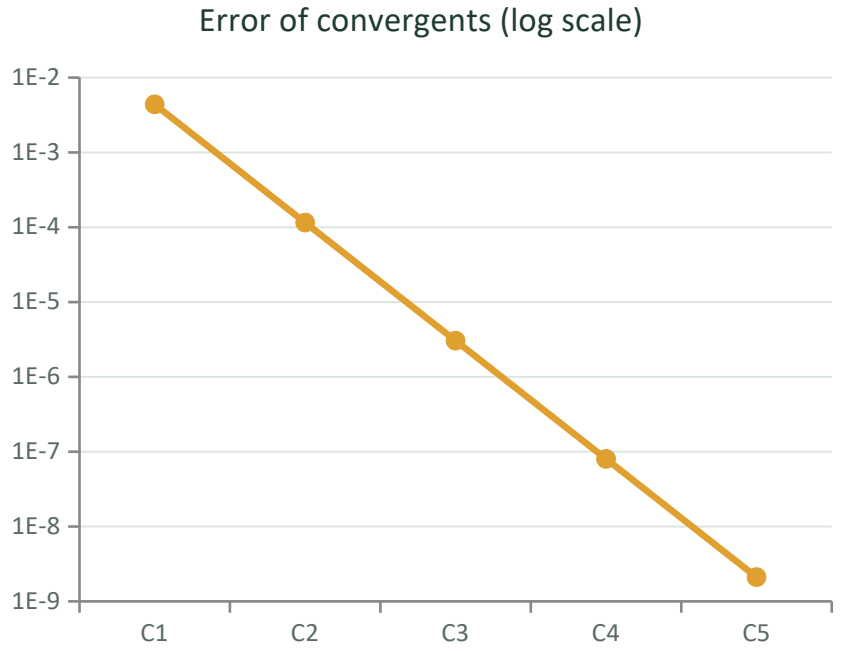


Result: $\sqrt{10} \approx 3.162277662$ Exact: $\sqrt{10} = 3.162277660168\ldots$
Absolute error: 2.11×10^{-9} obtained with only 5 divisions. Read the snake in the arrow direction: start at the deepest level t_5.

CF Ex. 5 · $\sqrt{10}$ · E3: convergents by forward recurrence

$A_k = b_k \cdot A_{k-1} + a_k \cdot A_{k-2}$ $B_k = b_k \cdot B_{k-1} + a_k \cdot B_{k-2}$ $C_k = A_k / B_k$ $(A_{-1} = 1, A_0 = b_0, B_{-1} = 0, B_0 = 1)$

k	a_k	b_k	A_k	B_k	$C_k = A_k/B_k$	$\lvert C_k - f \rvert$
0	—	3	3	1	3.000000000	0.1623
1	1	6	19	6	3.166666667	4.39×10^{-3}
2	1	6	117	37	3.162162162	1.15×10^{-4}
3	1	6	721	228	3.162280702	3.04×10^{-6}
4	1	6	4443	1405	3.162277580	8.01×10^{-8}
5	1	6	27379	8658	3.162277662	2.11×10^{-9}



$C_5 = 3.162277662 \equiv$ backward result ✓
≈ 8 correct decimals

Part I summary: results of the five continued-fraction examples

Quantity	Levels n	CF value C_n	Exact value	Absolute error
$e = e^1$	7	2.718283582	2.718281828459…	1.75×10^{-6}
$\tan 1$	5	1.557407407	1.557407724654…	3.17×10^{-7}
$\ln 2$	7	0.693152455	0.693147180559…	5.27×10^{-6}
$\arctan 1 = \pi/4$	6	0.785365854	0.785398163397…	3.23×10^{-5}
$\sqrt{10}$	5	3.162277662	3.162277660168…	2.11×10^{-9}

1 Same C_n, two routes

Backward scheme and forward recurrence give identical values — a built-in check.

2 Generated, not memorised

Every expansion came from its Taylor series (or from $x^2 = 10$) by the same G1–G3 recipe.

3 Alternating convergents

C_k oscillate around f when all $a_k > 0$: the true value is bracketed.

Part II · Transcendental equations by Rolle's sequence method

Use the zeros of f’ to count and separate all real roots of f(x) = 0, then refine each one

Rolle's theorem and the key corollary

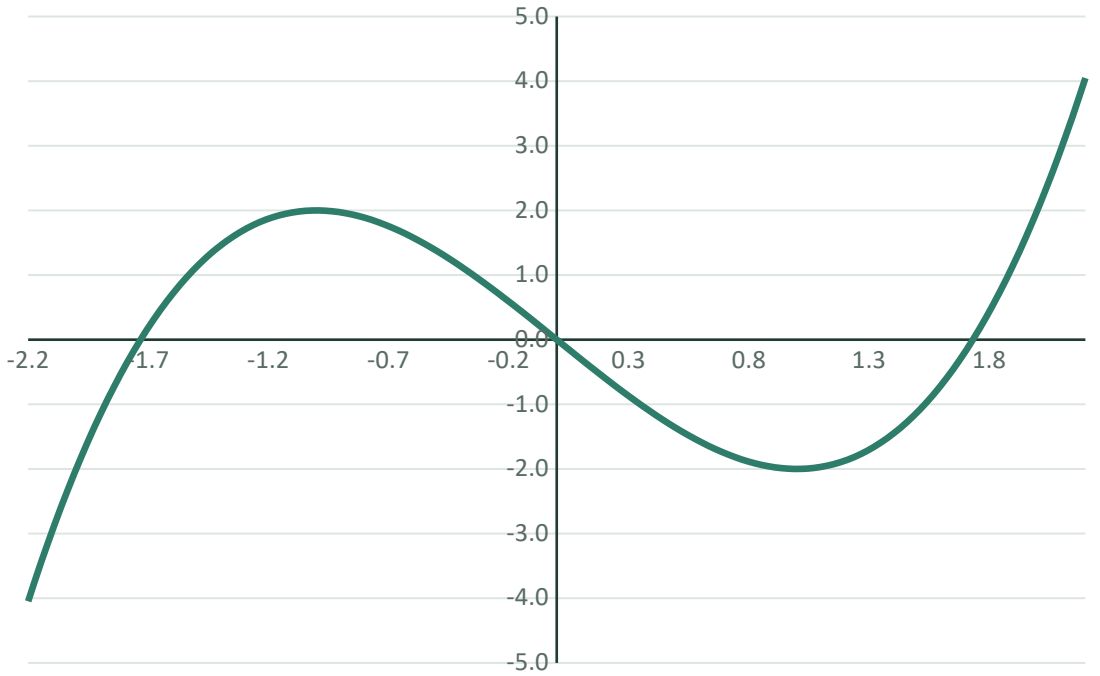
Rolle's theorem

If f is continuous on [a, b], differentiable on (a, b) and f(a) = f(b), then $\exists\, c \in (a, b)$ with $f'(c) = 0$.

Corollary (basis of the method)

- Between two consecutive zeros of f′ there is at most one zero of f.
- There is exactly one if f changes sign there; none otherwise.

f(x) = x³ − 3x : zeros of f′ at ±1 separate the 3 zeros of f



$f'(x) = 3x^2 - 3 = 0 \Rightarrow c = \pm 1$. Signs: $f(-\infty)$ −, $f(-1)$ +, $f(1)$ −, $f(+\infty)$ + ⇒ 3 sign changes ⇒ 3 real roots.

The algorithm of Rolle's sequence method

1 Domain

Fix $D = (\alpha, \beta)$ where f is continuous and differentiable.

2 Derivative

Compute $f'(x)$ and solve $f'(x) = 0$ on D (exactly or numerically).

3 Critical points

Order them: $\alpha < c_1 < c_2 < \ldots < c_k < \beta$.

4 Rolle's sequence

Evaluate $f(\alpha^+), f(c_1), \ldots, f(c_k), f(\beta^-)$ — limits at the ends.

5 Count sign changes

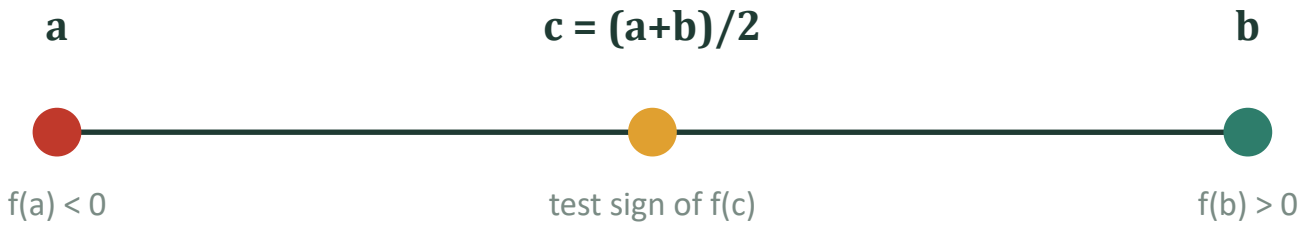
Each sign change ⇒ exactly one root in that interval.

6 Refine

Shrink to a finite bracket [a, b], then bisection (or Newton).

Remarks: if $f(c_i) = 0$, then c_i is a multiple root; number of real roots = number of sign changes in Rolle's sequence.

Refinement step: the bisection method



Keep the half where f changes sign:

- $f(a)\cdot f(c) < 0 \Rightarrow b \leftarrow c$
- otherwise $\Rightarrow a \leftarrow c$
- after n steps: $|\xi - c_n| \le (b-a) / 2^n$
- steps needed for tolerance ε: $n \ge \log_2((b-a)/\varepsilon)$

Reading the tables in the examples

k	iteration number
a, b	current bracket
c	midpoint (new estimate)
f(c)	value at the midpoint, coloured by sign

+ positive **– negative**

7 steps on a bracket of length 1 give an error ≤ 1/128 ≈ 0.0078; we also report the root to 7 decimals (≈ 25 steps).

Rolle Example 1 · $e^x - 3x = 0$: Steps 1–3 (setup)

1 **Domain**

$D = \mathbb{R}$ (f is continuous and differentiable)

2 **Derivative**

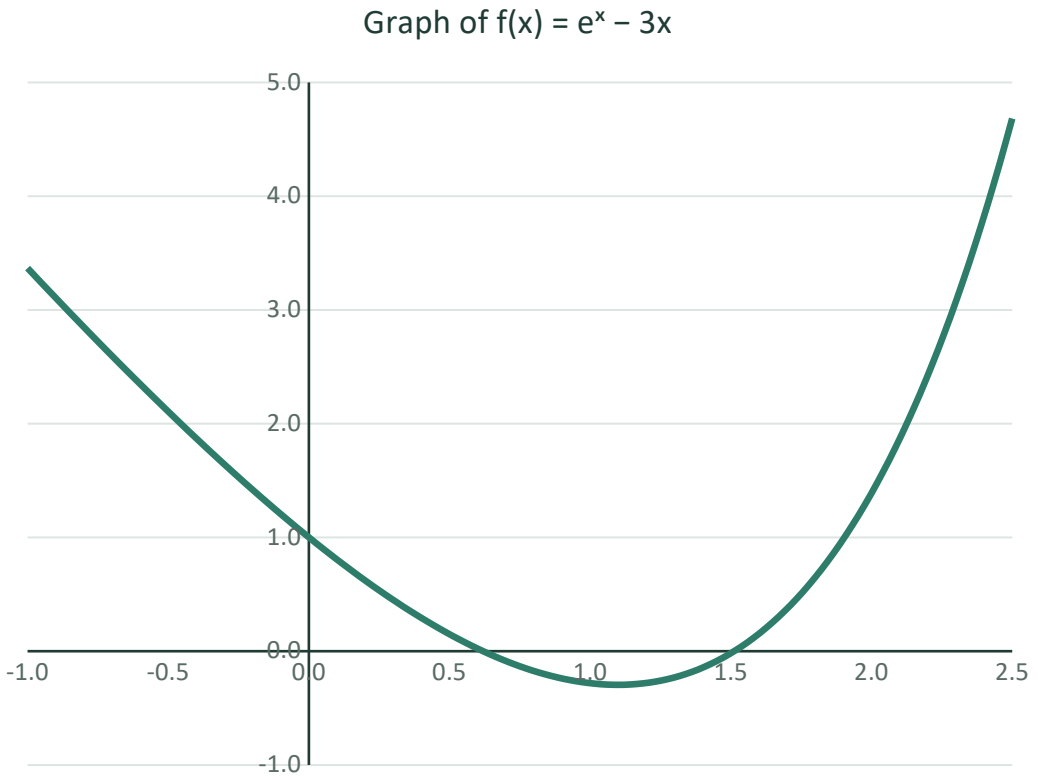
$f'(x) = e^x - 3$

3 **Critical points**

$e^x - 3 = 0 \Rightarrow c_1 = \ln 3 \approx 1.098612$

4 **Limits at the ends**

$x \to -\infty$: $f \to +\infty$; $x \to +\infty$: $f \to +\infty$



f decreases on $(-\infty, \ln 3)$ and increases on $(\ln 3, \infty)$; the minimum $f(\ln 3) = 3 - 3 \ln 3 < 0$.

Rolle Example 1 · Rolle's sequence and sign changes

Step 4 · Rolle's sequence for $f(x) = e^x - 3x$

x	$-\infty$	$\ln 3 \approx 1.0986$	$+\infty$
$f(x)$	$+\infty$	-0.295837	$+\infty$
sign	+	−	+



Step 5 · 2 sign changes ⇒ exactly 2 real roots — shrink each interval to a finite bracket

$\xi_1 \in (-\infty, \ln 3) \to (0, 1)$

$f(0) = 1 > 0,\ f(1) = e - 3 = -0.2817 < 0$

$\xi_2 \in (\ln 3, +\infty) \to (1, 2)$

$f(1) < 0$ (and $f < 0$ on $[1, \ln 3]$), $f(2) = e^2 - 6 = 1.3891 > 0$

Rolle Example 1 · Step 6: refine by bisection

ξ_1 on [0, 1]

k	a	b	c = (a+b)/2	f(c)
1	0.000000	1.000000	0.500000	**0.148721**
2	0.500000	1.000000	0.750000	**−0.133000**
3	0.500000	0.750000	0.625000	**−0.006754**
4	0.500000	0.625000	0.562500	**0.067555**
5	0.562500	0.625000	0.593750	**0.029516**
6	0.593750	0.625000	0.609375	**0.011156**
7	0.609375	0.625000	0.617188	**0.002145**

$\xi_1 \approx c_7 = 0.617188$, $|\text{error}| \le 1.0/2^7 = 0.0078$
Continuing the bisection: $\xi_1 = 0.6190613$

ξ_2 on [1, 2]

k	a	b	c = (a+b)/2	f(c)
1	1.000000	2.000000	1.500000	**−0.018311**
2	1.500000	2.000000	1.750000	**0.504603**
3	1.500000	1.750000	1.625000	**0.203419**
4	1.500000	1.625000	1.562500	**0.083233**
5	1.500000	1.562500	1.531250	**0.030203**
6	1.500000	1.531250	1.515625	**0.005390**
7	1.500000	1.515625	1.507813	**−0.006598**

$\xi_2 \approx c_7 = 1.507813$, $|\text{error}| \le 1.0/2^7 = 0.0078$
Continuing the bisection: $\xi_2 = 1.5121346$

Rolle Example 2 · ln x – x + 2 = 0: Steps 1–3 (setup)

1 **Domain**

$D = (0, +\infty)$ (because of $\ln x$)

2 **Derivative**

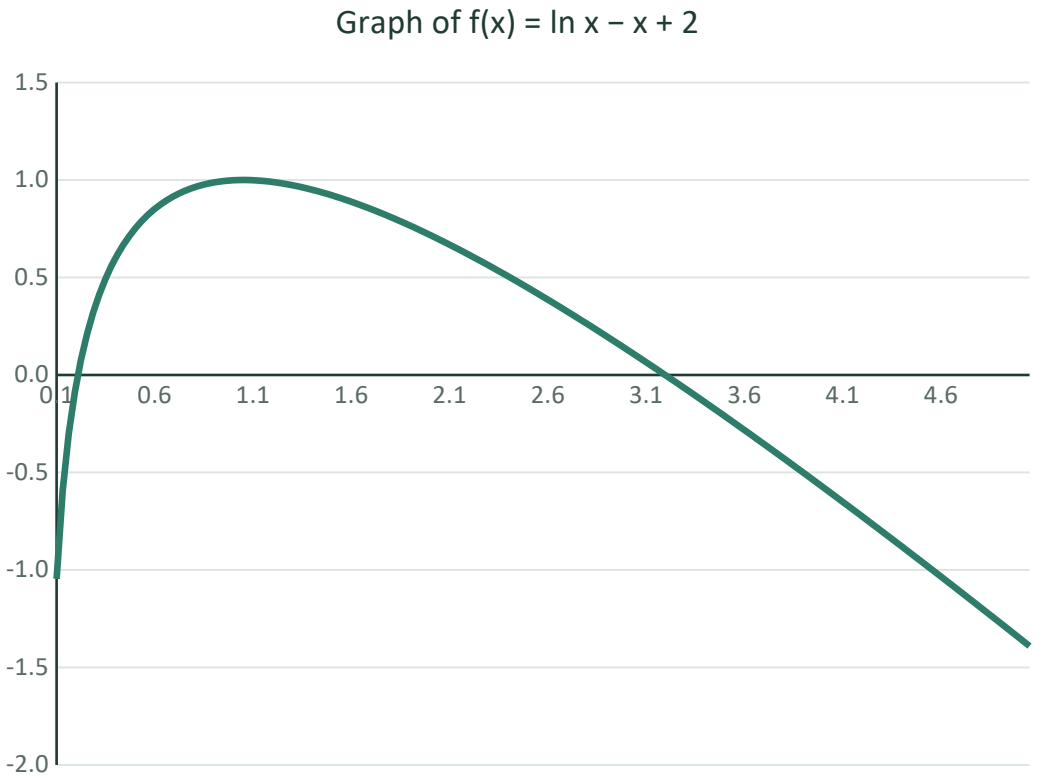
$f'(x) = 1/x - 1 = (1 - x)/x$

3 **Critical points**

$f'(x) = 0 \Rightarrow c_1 = 1$; $f(1) = 0 - 1 + 2 = 1$

4 **Limits at the ends**

$x \to 0^+: f \to -\infty$; $x \to +\infty: f \to -\infty$



f increases on (0, 1) and decreases on (1, ∞); the maximum f(1) = 1 > 0.

Rolle Example 2 · Rolle's sequence and sign changes

Step 4 · Rolle's sequence for f(x) = ln x − x + 2

x	0^+	1	$+\infty$
f(x)	$-\infty$	1.000000	$-\infty$
sign	−	+	−



Step 5 · 2 sign changes ⇒ exactly 2 real roots — shrink each interval to a finite bracket

$\xi_1 \in (0, 1) \to (0.1, 0.2)$

$f(0.1) = -0.4026 < 0,\ f(0.2) = 0.1906 > 0$

$\xi_2 \in (1, +\infty) \to (3, 4)$

$f(3) = 0.0986 > 0,\ f(4) = -0.6137 < 0$

Rolle Example 2 · Step 6: refine by bisection

ξ_1 on [0.1, 0.2]

k	a	b	c = (a+b)/2	f(c)
1	0.100000	0.200000	0.150000	**−0.047120**
2	0.150000	0.200000	0.175000	**0.082031**
3	0.150000	0.175000	0.162500	**0.020423**
4	0.150000	0.162500	0.156250	**−0.012548**
5	0.156250	0.162500	0.159375	**0.004130**
6	0.156250	0.159375	0.157813	**−0.004160**
7	0.157813	0.159375	0.158594	**−0.000003**

$\xi_1 \approx c_7 = 0.158594$, $|\text{error}| \le 0.1/2^7 = 0.0008$
Continuing the bisection: $\xi_1 = 0.1585943$

ξ_2 on [3, 4]

k	a	b	c = (a+b)/2	f(c)
1	3.000000	4.000000	3.500000	**−0.247237**
2	3.000000	3.500000	3.250000	**−0.071345**
3	3.000000	3.250000	3.125000	**0.014434**
4	3.125000	3.250000	3.187500	**−0.028263**
5	3.125000	3.187500	3.156250	**−0.006865**
6	3.125000	3.156250	3.140625	**0.003797**
7	3.140625	3.156250	3.148438	**−0.001531**

$\xi_2 \approx c_7 = 3.148438$, $|\text{error}| \le 1.0/2^7 = 0.0078$
Continuing the bisection: $\xi_2 = 3.1461932$

Rolle Example 3 · x – 2 sin x = 0: Steps 1–3 (setup)

1 **Domain and bounding**

$D = \mathbb{R}$; for $|x| > 2$: $|2 \sin x| \le 2 < |x| \Rightarrow$ no roots, so work on $[-2, 2]$

2 **Derivative**

$f'(x) = 1 - 2 \cos x$

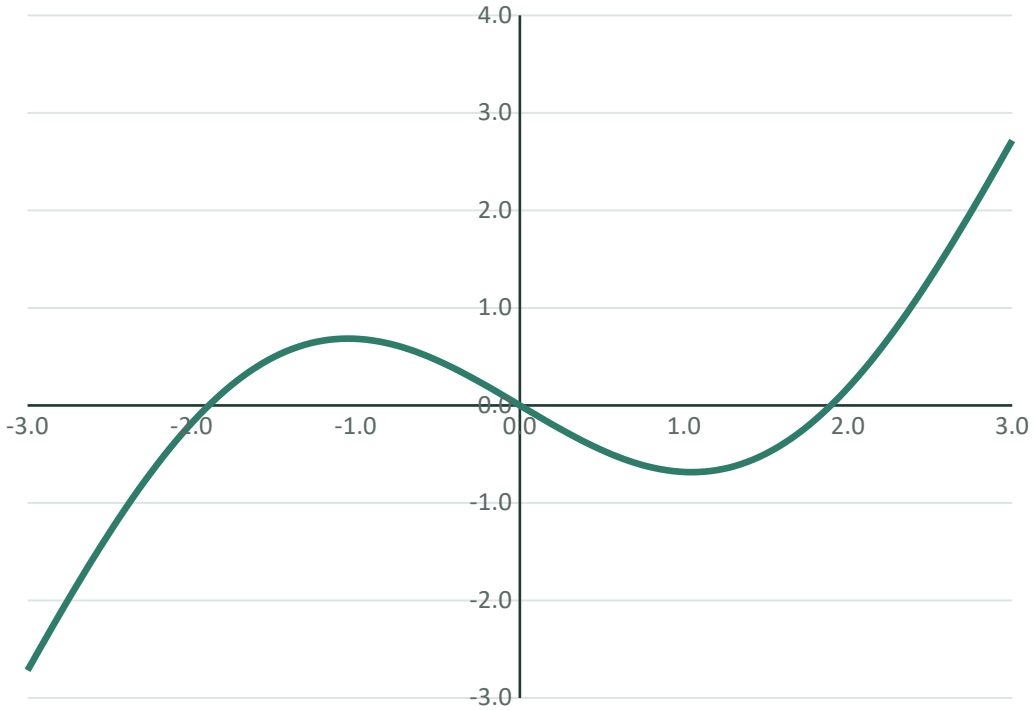
3 **Critical points in [–2, 2]**

$\cos x = 1/2 \Rightarrow c_1 = -\pi/3,\ c_2 = \pi/3\ (\pm 1.047198)$

4 **Values at the ends**

$f(-2) = -0.181405,\ \ f(2) = 0.181405$

Graph of f(x) = x − 2 sin x



f is odd: f(−x) = −f(x). So roots come in pairs ±ξ, and x = 0 is a root.

Rolle Example 3 · Rolle's sequence and sign changes

Step 4 · Rolle's sequence for f(x) = x − 2 sin x

x	−2	−π/3	π/3	2
f(x)	−0.181405	0.684853	−0.684853	0.181405
sign	−	+	−	+



Step 5 · 3 sign changes ⇒ exactly 3 real roots — shrink each interval to a finite bracket

$\xi_1 \in (-2, -\pi/3) \to (-2, -1.5)$

$f(-2) < 0,\ f(-1.5) = 0.4950 > 0$

$\xi_2 \in (-\pi/3, \pi/3)$

$f(0) = 0 \Rightarrow \xi_2 = 0$ exactly

$\xi_3 \in (\pi/3, 2) \to (1.5, 2)$

$f(1.5) = -0.4950,\ f(2) = 0.1814;\ \xi_1 = -\xi_3$

Rolle Example 3 · Step 6: refine by bisection

ξ_3 on [1.5, 2]

k	a	b	c = (a+b)/2	f(c)
1	1.500000	2.000000	1.750000	**−0.217972**
2	1.750000	2.000000	1.875000	**−0.033172**
3	1.875000	2.000000	1.937500	**0.070471**
4	1.875000	1.937500	1.906250	**0.017728**
5	1.875000	1.906250	1.890625	**−0.007954**
6	1.890625	1.906250	1.898438	**0.004829**
7	1.890625	1.898438	1.894531	**−0.001577**

$\xi_3 \approx c_7 = 1.894531$, $|\text{error}| \le 0.5/2^7 = 0.0039$
Continuing the bisection: $\xi_3 = 1.8954943$

ξ_1 on [−2, −1.5]

k	a	b	c = (a+b)/2	f(c)
1	−2.000000	−1.500000	−1.750000	**0.217972**
2	−2.000000	−1.750000	−1.875000	**0.033172**
3	−2.000000	−1.875000	−1.937500	**−0.070471**
4	−1.937500	−1.875000	−1.906250	**−0.017728**
5	−1.906250	−1.875000	−1.890625	**0.007954**
6	−1.906250	−1.890625	−1.898438	**−0.004829**
7	−1.898438	−1.890625	−1.894531	**0.001577**

$\xi_1 \approx c_7 = -1.894531$, $|\text{error}| \le 0.5/2^7 = 0.0039$
Continuing the bisection: $\xi_1 = -1.8954943$

Symmetry check: the two tables are mirror images (f odd) $\Rightarrow \xi_1 = -\xi_3$. Together with $\xi_2 = 0$ the equation has exactly 3 real roots.

Rolle Example 4 · $x^2 \cdot e^{-x} - 0.1 = 0$: Steps 1–3 (setup)

1 **Domain**

$D = \mathbb{R}$

2 **Derivative**

$f'(x) = 2x \cdot e^{-x} - x^2 \cdot e^{-x} = x(2 - x) \cdot e^{-x}$

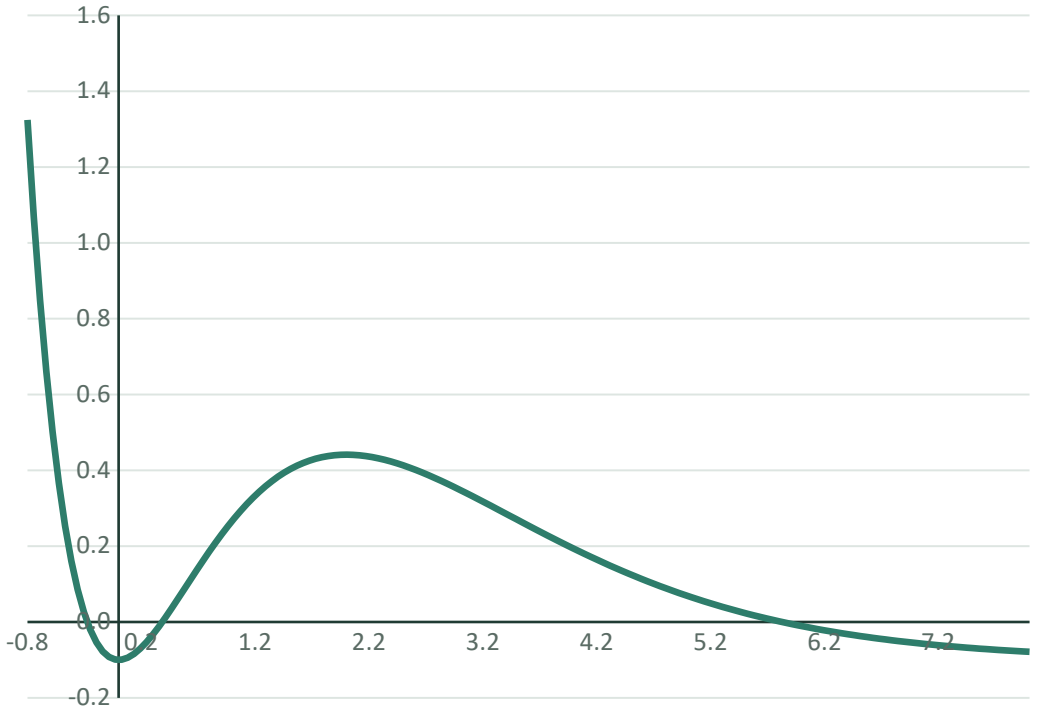
3 **Critical points**

$c_1 = 0$ (min), $c_2 = 2$ (max)

4 **Limits at the ends**

$x \to -\infty$: $f \to +\infty$; $x \to +\infty$: $f \to -0.1$

Graph of $f(x) = x^2 e^{-x} - 0.1$



Rolle values: $f(0) = -0.1$ and $f(2) = 4/e^2 - 0.1 = 0.441341$. Note $f(+\infty) = -0.1$ is a finite limit, still negative.

Rolle Example 4 · Rolle's sequence and sign changes

Step 4 · Rolle's sequence for $f(x) = x^2e^{-x} - 0.1$

x	$-\infty$	0	2	$+\infty$
f(x)	$+\infty$	-0.100000	0.441341	-0.1
sign	+	−	+	−



Step 5 · 3 sign changes ⇒ exactly 3 real roots — shrink each interval to a finite bracket

$\xi_1 \in (-\infty, 0) \to (-1, 0)$

$f(-1) = e - 0.1 = 2.6183 > 0,\ f(0) = -0.1$

$\xi_2 \in (0, 2) \to (0, 1)$

$f(0) = -0.1 < 0,\ f(1) = 1/e - 0.1 = 0.2679 > 0$

$\xi_3 \in (2, +\infty) \to (5, 6)$

$f(5) = 0.0684 > 0,\ f(6) = -0.0108 < 0$

Rolle Example 4 · Step 6: refine by bisection (part 1)

ξ_1 on [−1, 0]

k	a	b	c = (a+b)/2	f(c)
1	−1.000000	0.000000	−0.500000	**0.312180**
2	−0.500000	0.000000	−0.250000	**−0.019748**
3	−0.500000	−0.250000	−0.375000	**0.104608**
4	−0.375000	−0.250000	−0.312500	**0.033480**
5	−0.312500	−0.250000	−0.281250	**0.004793**
6	−0.281250	−0.250000	−0.265625	**−0.007977**
7	−0.281250	−0.265625	−0.273438	**−0.001719**

$\xi_1 \approx c_7 = -0.273438$, $|\text{error}| \le 1.0/2^7 = 0.0078$
Continuing the bisection: $\xi_1 = -0.2755303$

ξ_2 on [0, 1]

k	a	b	c = (a+b)/2	f(c)
1	0.000000	1.000000	0.500000	**0.051633**
2	0.000000	0.500000	0.250000	**−0.051325**
3	0.250000	0.500000	0.375000	**−0.003350**
4	0.375000	0.500000	0.437500	**0.023581**
5	0.375000	0.437500	0.406250	**0.009940**
6	0.375000	0.406250	0.390625	**0.003246**
7	0.375000	0.390625	0.382813	**−0.000065**

$\xi_2 \approx c_7 = 0.382813$, $|\text{error}| \le 1.0/2^7 = 0.0078$
Continuing the bisection: $\xi_2 = 0.3829658$

Rolle Example 4 · Step 6: refine by bisection (part 2)

ξ_3 on [5, 6]

k	a	b	c = (a+b)/2	f(c)
1	5.000000	6.000000	5.500000	0.023625
2	5.500000	6.000000	5.750000	0.005231
3	5.750000	6.000000	5.875000	−0.003053
4	5.750000	5.875000	5.812500	0.001016
5	5.812500	5.875000	5.843750	−0.001037
6	5.812500	5.843750	5.828125	−0.000015
7	5.812500	5.828125	5.820313	0.000499

$\xi_3 \approx c_7 = 5.820313$, $|\text{error}| \le 1.0/2^7 = 0.0078$
Continuing the bisection: $\xi_3 = 5.8278978$

All three roots

$\xi_1 = -0.2755303$

$\xi_2 = 0.3829658$

$\xi_3 = 5.8278978$

Check: f(ξ) ≈ 0 for each root, and the number of roots equals the number of sign changes in Rolle's sequence.

Rolle Example $5 \cdot 2^x - x^2 = 0$: Steps 1–3 (setup)

1 **Domain**

$D = \mathbb{R}$

2 **Derivative**

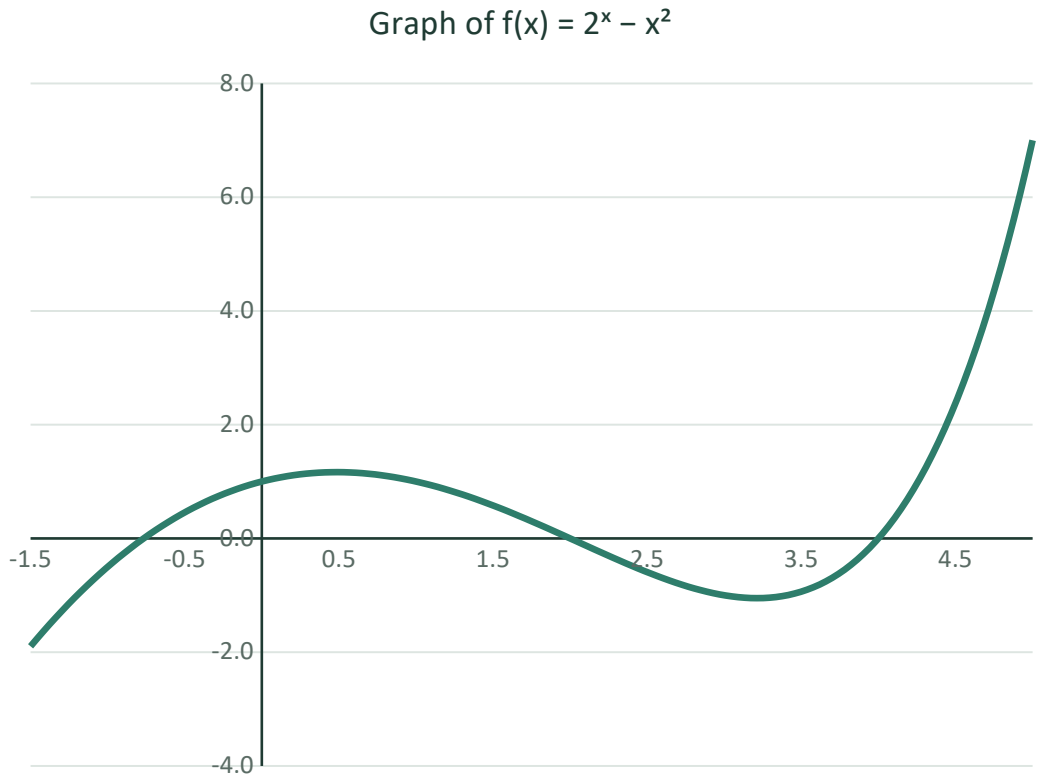
$f'(x) = 2^x \cdot \ln 2 - 2x$

3 **Problem**

$f'(x) = 0$ is itself transcendental ⇒ apply Rolle's method to f' first

4 **Limits at the ends**

$x \to -\infty: f \to -\infty; \quad x \to +\infty: f \to +\infty$



Two exact roots can be guessed: $2^2 = 4 = 2^2$ and $2^4 = 16 = 4^2$. Is there a third one? Rolle's sequence will decide.

Rolle Example 5 · Level 1: Rolle's sequence for f′ (via f″)

$f''(x) = 2^x \cdot \ln^2 2 - 2 = 0 \Rightarrow x^* = \log_2(2 / \ln^2 2) \approx 2.057533$ (single zero of f″, solvable exactly)

Rolle's sequence for $f'(x) = 2^x \ln 2 - 2x$

x	$-\infty$	$x^* \approx 2.0575$	$+\infty$
f(x)	$+\infty$	-1.229675	$+\infty$
sign	+	−	+

ξ_1 ξ_2

$c_1 \in (-\infty, x^*) \to (0, 1)$

$f'(0) = \ln 2 = 0.6931 > 0$, $f'(1) = 2 \ln 2 - 2 = -0.6137 < 0$
bisection $\Rightarrow c_1 = 0.485090$

$c_2 \in (x^*, +\infty) \to (3, 4)$

$f'(3) = 8 \ln 2 - 6 = -0.4548 < 0$, $f'(4) = 16 \ln 2 - 8 = 3.0904 > 0$
bisection $\Rightarrow c_2 = 3.212433$

Rolle Example 5 · Rolle's sequence and sign changes

Step 4 · Rolle's sequence for $f(x) = 2^x - x^2$

x	$-\infty$	$c_1 \approx 0.4851$	$c_2 \approx 3.2124$	$+\infty$
f(x)	$-\infty$	1.164361	−1.050602	$+\infty$
sign	−	+	−	+



Step 5 · 3 sign changes ⇒ exactly 3 real roots — shrink each interval to a finite bracket

$\xi_1 \in (-\infty, c_1) \to (-1, 0)$

$f(-1) = 0.5 - 1 = -0.5 < 0,\ f(0) = 1 > 0$

$\xi_2 \in (c_1, c_2)$

$f(2) = 4 - 4 = 0 \Rightarrow \xi_2 = 2$ exactly

$\xi_3 \in (c_2, +\infty)$

$f(4) = 16 - 16 = 0 \Rightarrow \xi_3 = 4$ exactly

Rolle Example 5 · Step 6: refine ξ_1 by bisection, verify ξ_2 and ξ_3

ξ_1 on [−1, 0]

k	a	b	c = (a+b)/2	f(c)
1	−1.000000	0.000000	−0.500000	**0.457107**
2	−1.000000	−0.500000	−0.750000	**0.032104**
3	−1.000000	−0.750000	−0.875000	**−0.220371**
4	−0.875000	−0.750000	−0.812500	**−0.090762**
5	−0.812500	−0.750000	−0.781250	**−0.028489**
6	−0.781250	−0.750000	−0.765625	**0.002017**
7	−0.781250	−0.765625	−0.773438	**−0.013184**

$\xi_1 \approx c_7 = -0.773438$, $|\text{error}| \le 1.0/2^7 = 0.0078$
Continuing the bisection: $\xi_1 = -0.7666647$

All three roots

$\xi_1 = -0.7666647$

$\xi_2 = 2 \quad (2^2 = 4 = 2^2)$

$\xi_3 = 4 \quad (2^4 = 16 = 4^2)$

Without Rolle's sequence the negative root is easy to miss; the method also proves there is no fourth root.

Part II summary: roots found by Rolle's sequence method

Equation	# real roots	Roots (7 decimals)
$e^x - 3x = 0$	2	$\xi_1 = 0.6190613;\ \ \xi_2 = 1.5121346$
$\ln x - x + 2 = 0$	2	$\xi_1 = 0.1585943;\ \ \xi_2 = 3.1461932$
$x - 2\sin x = 0$	3	$\xi_1 = -1.8954943;\ \ \xi_2 = 0;\ \ \xi_3 = 1.8954943$
$x^2 \cdot e^{-x} - 0.1 = 0$	3	$\xi_1 = -0.2755303;\ \ \xi_2 = 0.3829658;\ \ \xi_3 = 5.8278978$
$2^x - x^2 = 0$	3	$\xi_1 = -0.7666647;\ \ \xi_2 = 2;\ \ \xi_3 = 4$

1 Count first, compute later

Sign changes give the exact number of real roots before any iteration.

2 Finite limits count too

Ex. 4: $f(+\infty) = -0.1$ is enough to decide the sign at the end.

3 Nested Rolle

If $f' = 0$ is transcendental, apply the method to f' (Ex. 5).

Key takeaways

Continued fractions: few divisions, many digits — evaluate bottom-up or by the A_k, B_k recurrence.

Rolle's sequence: zeros of f′ split the axis; sign changes count and isolate every real root; bisection finishes the job.